Competing for Quality: Coordination Failure in Dairy Genetics Markets

Abstract

We explore how a new information technology, genomic selection, exacerbates a coordination failure in the market for dairy cattle genetics in the United States. Genomic selection allows cattle breeders to test new cattle as soon as they are born for the presence of desirable, productive traits and more than double their pace of genetic improvement in milk production. At the same time, the rate of inbreeding, the extent to which dairy bulls are related to one another, greatly increased after the introduction of genomic testing. In a market where firms must produce the highest quality genetics to capture market share, we hypothesize that genomic testing increased incentives to inbreed, an action which leads to welfare losses in the industry by dampening genetic improvement in the future. Using a data set of bulls marketed in the US from 2000 to 2020, we test whether inbreeding rates increased where the returns to inbreeding were highest: dairy bull families that were popular before genomic testing. We employ a differences-in-differences and matching approach and find that inbreeding increased about 50% in the already popular families after the introduction of genomic testing, confirming our hypothesis. Together with our conceptual model, our results suggests that these actions of animal breeders may be harming productivity growth in the industry in the long run.

1 Introduction

The US plays a crucial role in the global dairy sector as one of the world's largest suppliers of dairy bull genetics. In 2022, the United States alone exported $295 million of bovine semen, representing 47% of the world's exports in value terms (Figure 1). For about six decades, milk production per cow has consistently increased in the US herd, and it is estimated that more than half of this growth can be attributed to improvements in cattle genetics (Council on Dairy Cattle Breeding, 2025). Thus, the efforts of US breeders have likely had substantial impacts on the global dairy industry.

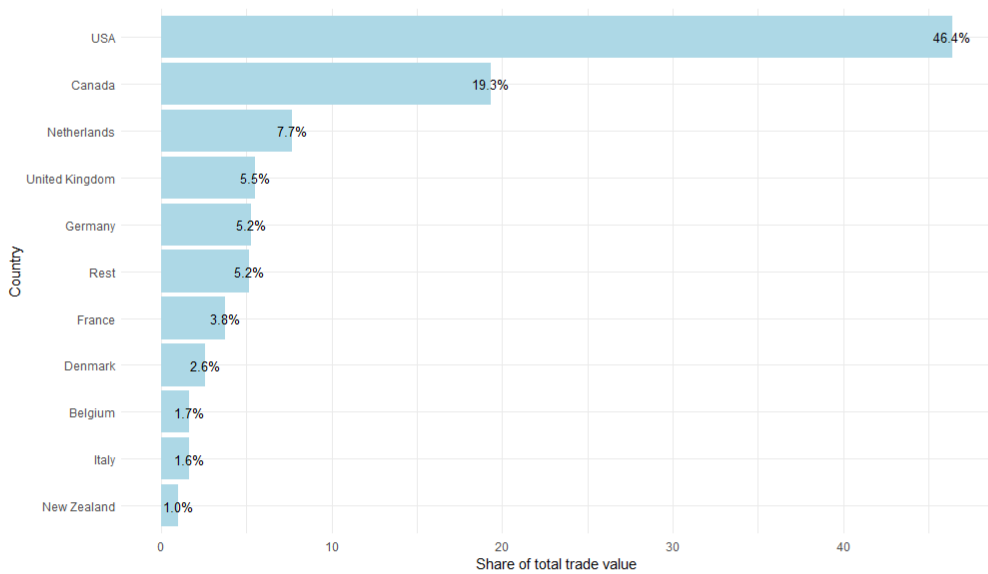

Figure 1: Share of trade Value of Bovine Semen exports by country (2019)
Source: UN Comtrade database

In the modern dairy industry, dairy breeders improve cow milk yield by identifying which bulls have the most productive female offspring and then selling these semen to dairy farmers looking to improve their genetics. The speed of this genetic improvement has previously been limited by biological factors. Determining whether a bull had good milk production ability took at least five years since the bull had to have offspring born from their use on dairy farms, and those offspring would need to take two years to grow to production age.

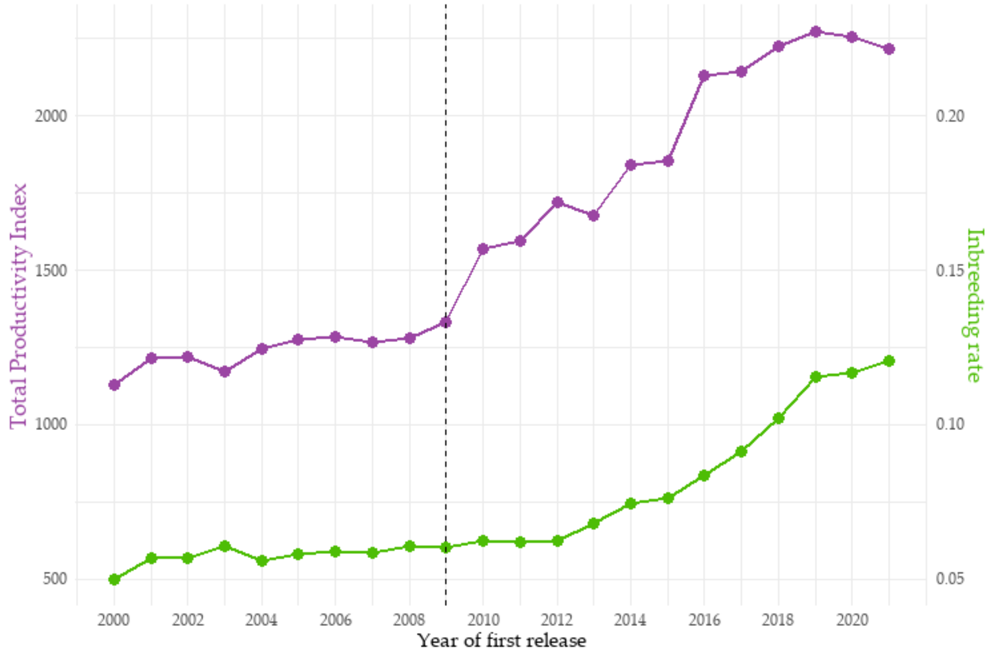

Figure 2: Dairy Bull Productivity versus Inbreeding Rate
Source: National Association of Animal Breeders (TPI)
and Canadian Dairy Network (Inbreeding rate)

However, these limits became less binding after the advent of an innovation in testing technology: genomic testing. Introduced in 2008, this technological innovation allows breeders to link each animal's genes to productive outcomes such as milk yield and fat yield. The practical upshot is that determining a dairy bull's quality can now be done at birth rather than in 5 years. There are two important, documented impacts of genomic testing on the dairy bull market. First, the pace of genetic improvement has more than doubled in dairy cattle since breeders can find productive bulls at much younger ages and breed new generations even quickly. Second, dairy companies have more than doubled the number of bulls for sale on the market since young bulls produce less semen than older animals; therefore, more animals are required to meet revenue goals.

The introduction of genomic testing has coincided with a less desirable outcome: an increase in inbreeding, the degree to which a dairy bull is related to the rest of the dairy cattle population. Figure 2 shows the average inbreeding rate of dairy bulls on the market alongside the total productivity index (TPI), a measure of overall genetic performance value[1].

Excessive inbreeding is an example of a coordination failure in the cattle genetics industry. Dairy producers are incentivized to increase the quality of their bulls by either inbreeding or outcrossing. Inbreeding captures a larger market share from their competitors by producing more quality, but doing so depletes their own stock of genetic diversity. By outcrossing and using another company's genetic stock, they can replenish their own genetic diversity but gain less market share because their quality improvement will be lower. These dynamics incentivize dairy breeders to inbreed more than would be beneficial to the industry because it is costly to outcross when the rest of the industry inbreeds. Because their own inbreeding makes outcrossing less valuable for firms that may want to use those lines, individual breeders also do not fully consider the costs of their own inbreeding decision. The result is that inbreeding is done more often in the industry than would be socially optimal.

Excessive inbreeding is harmful to the industry because it slows growth in genetic quality and can cause costs down the road that can seriously harm the dairy industry. In one case, breeding excessively from a famous dairy bull "Pawnee Farm Arlinda Chief" led to more than 500,000 spontaneous abortions and nearly 420 million dollars in losses because of an undesirable, recessive trait that was passed to his offspring (Adams et al., 2016). Given the US's role in the global cattle genetics market, reducing the diversity of the dairy cattle gene pool could damage livestock markets worldwide.

In this paper, we examine whether the introduction of genomic testing increased inbreeding by increasing the returns to inbreeding relative to outcrossing for bulls who were well known in the industry. Genomic testing significantly increased the returns to inbreeding on these lines because, being the most widely used, the information gain from sequencing these lines was highest. Genomic selection also made it easier to detect productive genes that they want future offspring to inherit through inbreeding. Since

[1] TPI is an economic value-weighted average of a set of m genetic traits: $H = \sum_{j=1}^{m} p_j g_j$, where p_j and g_j are the economic values and breeding values of trait j (Melton et al., 1979)

well-known bulls are most likely to also be high quality, we construct a set of counterfactual bulls that are comparable to the popular genetic families in terms of productivity using propensity score matching. Using a difference-in-difference approach, we then examine inbreeding rates in popular families relative to our counterfactual families before and after the introduction of genomic testing in 2010. While the counterfactual of an industry without genomic testing is not observable, we focus on identifying the increase in inbreeding relative to bull lines where genomic testing would increase the returns to inbreeding.

We find that popular dairy bull lines became more inbred after the introduction of genomic testing than our counterfactual. The inbreeding rate of the most popular lines increased by 1.2-3.5 percentage points relative to our control group after 2010, about a 50% increase compared to the rates pre-2010. We also find no statistically significant difference in inbreeding rates across our groups before genomic testing was introduced. Using a rough estimate of the costs of inbreeding from the breeding association Holstein USA, a back-of-the-envelope calculation implies that, on net, the benefits of inbreeding outweighed the costs for this group: increased inbreeding led to approximately USD 1.6-4.2 billion in losses from 2012 to 2019, while the benefits net of inbreeding cost totaled USD 2.7-6 billion.

This article fills a critical gap in the literature on livestock markets and the behavior of animal breeders. Few economics papers analyze the strategic actions of animal breeders, and even fewer analyze how their actions may impose costs on the industry. Early modeling of the animal breeder's decision dates back to studies such as Ladd and Gibson (1978), Kerr (1984), and Melton et al. (1994), which use hedonic regression and linear programming to determine the implicit market value of different livestock traits. Richards and Jeffrey (1996) and Schroeder et al. (1992) apply this method to dairy bulls to determine the economic value of each dairy bull trait implied by the results of a hedonic regression. Thus far, the economics literature has not recognized that animal breeders' decisions may lead to worse outcomes for the entire sector.

However, a related literature in agricultural economics has explored similar coordination problems among farmers controlling pests or disease. In the short run, farmers are incentivized to forgo the costs of controlling a future pest or disease because of a strong incentive to free-ride off of the efforts of others. This has been documented in the case of planting Bt-resistant crops (Brown, 2018, Epanchin-Niell and Wilen, 2015, Livingston et al., 2004, Wechsler and Smith, 2018) and managing citrus greening Lence and Singerman (2023).

Even in the absence of explicit externalities, individual, profit-maximizing firms can find themselves in situations where product quality would be improved by coordination despite not being individually rational. Our paper is the first to explore such a coordination problem in the market for dairy cattle genetics. Breeders are incentivized in the short run to increase inbreeding by breeding cattle from popular lines, which may boost genetic improvement and capture more market share. However, inbreeding expends a type of resource, genetic diversity, that is needed to keep genetic improvement growing over time. Yet, it is individually rational to breed more than is socially optimal to maintain their market share by producing the bulls with the highest quality (Cole, 2019). The welfare loss occurs because maintaining a firm's genetic diversity is beneficial to the entire industry since other firms can use it to restore their own diversity. However, an individual firm does not consider this benefit, leading to a slower rate of genetic improvement in the long run.

Before its introduction, it was theorized that genomic testing would **decrease** inbreeding by allowing breeders to find productive dairy bulls outside known genetic lines more easily (Daetwyler et al., 2007). Instead, we find that the average inbreeding coefficient of dairy bulls increased along with dairy bull productivity. In our conceptual framework, we explore how genomic testing increases the returns to inbreeding and likely decreases the amount of genetic diversity in the entire industry. We provide evidence of this theory by comparing the inbreeding rates of groups of bulls that are similar in genetic traits but different in popularity before and after genomic testing. If genomic testing led to an increase in the discovery of less popular lines, we should expect no difference in the inbreeding rates of these groups since animal breeders would have selected them only on the basis of their genetic traits. Instead, we find that genetic lines already popular before genomic testing experienced an increase in the average inbreeding rate, leading to damage to the dairy sector.

If inbreeding continues to increase in the industry, even more damages might occur. Studies already estimate that the Holstein breed has an unsustainable level of inbreeding, having an “effective population” of between 43 and 66; it is commonly assumed that 50 is a critical level under which the breed will lose resilience (Makanjuola et al., 2020). Moreover, the lower the genetic diversity, the greater the risk there is of unknown recessive traits causing significant damage to genetic lines (Cole and VanRaden, 2011, Lozada-Soto et al., 2024). At its worst, the population can reach the maximum possible increase in genetic trait values and lead to a stalling of genetic improvement. Future work in economics should seek to understand how policy might be used to address market frictions and incentivize breeders to outcross to ameliorate the impacts of inbreeding.

The paper is structured as follows: Section 2 presents background and a conceptual model exploring how introducing genomic selection can lead to higher inbreeding rates in subsequent generations of dairy bulls. Section 3 explains the data sources and shows descriptive statistics. Section 4 outlines the identification strategy and methodological framework. In section 5, we explain the results before doing a cost-benefit analysis in section A. We conclude in section 6 and discuss the implications for the future of the US dairy industry.

2 Conceptual Framework

2.1 The Coordination Problem in Dairy Breeding

To describe the dynamics of the dairy industry, we present a simplified theoretical model that describes the coordination problem inherent to dairy cow breeding. In the modern dairy industry, breeding and selecting new bulls for sale has become almost exclusively the job of dairy breeders. In contrast, dairy farmers are largely uninvolved with the breeding process and buy their genetics from these companies instead of producing it themselves or selling genetics. Thus, we model the dairy breeder as a single breeding enterprise and not a farm also engaged in milk production.

In our theoretical formulation, there is a market made up of N firms and each firm i in time t produces quality q_{it} to earn a share of dairy industry revenue $R(\bar{Q})$, where $\bar{Q}$ is the average quality produced by all breeders. We assume that the dairy bull market has poor differentiation such that dairy farmers roughly treat every dairy bull as equivalent, and thus only pay attention to the amount of q_{it} the firm produces. This creates a zero-sum market environment, as a firm i that produces less quality than its competitors (q_{-it}) will receive a lower share $s(q_{it}, q_{-it})$ of the market revenue $R(\bar{Q})$.

Firms have two options (a_{it}) to increase quality: inbreed (I) or outcross (O). The returns to these options depend either on the genetic diversity of the firm, d_{it}, or the genetic diversity of the other firms, d_{-it}. The actions in time t will determine the quality in time $t+1$: $q_{it+1} = q_{it} + f(a_{it}, d_{it}, d_{-it})$, where f is:

$$f(a_{it}, d_{it}, d_{-it}) = \begin{cases} \gamma_I \cdot d_{it} & \text{if } a_{it} = I \\ \gamma_O \cdot d_{-it} & \text{if } a_{it} = O. \end{cases}$$

We assume that $\gamma_I > \gamma_O$ because inbreeding extracts more of the value from genetic diversity than outcrossing extracts from someone else's line. However, we also assume that choosing $a_{it} = I$ *depletes* d_{it} while $a_{it} = O$ *replenishes* d_{it} by adding genetic material from other lines. We assume the dynamics of d_{it} as:

$$d_{it+1} = \begin{cases} \delta \cdot d_{it} & \text{if } a_{it} = I \\ \min\{(1+\rho)d_{it}, 1\} & \text{if } a_{it} = O \end{cases}$$

where $\delta \in (0,1)$ is a depreciation rate and $\rho > 0$ is a replenishment rate. For simplicity, we assume that there is some upper limit to d_{it} that we can normalize to 1. The firm's tradeoff is whether to inbreed to increase q_{it+1}, and thus their revenue share next period $s(q_{it+1}, q_{-i,t+1})R(\bar{Q})$, at the expense of their stock d_{it}. Outcrossing would restore their genetic stock but might grant market share to their competitors because they will gain less q_{it}.

With the most basic cost function, one where a firm pays α_I to inbreed and α_O to outcross, the firm's maximization problem as a Bellman equation would be:

$$\begin{aligned} max_{a\in(I,O)} \quad & V(a,q,d) = R(\bar{Q})s(q,q_{-i}) - C(a) + \beta V(q'(a), d'(a)) \\ \text{s.}t. \quad & C(a) = \alpha_I\{a = I\} + \alpha_O\{a = O\} \\ & \text{rules of motion for } d \text{ an } q \end{aligned}$$

where $\beta \in [0,1)$ is the discount rate.

In many cases, the optimal action for a breeder would be to inbreed regardless of what the other firms do. Assuming only two players and two periods, this can be illustrated as a 2x2 game (assuming symmetric d and q):

		Breeder 2	
		Inbreed	Outcross
Breeder 1	Inbreed	$R(\bar{Q}_{II})/2$, $R(\bar{Q}_{II})/2$	$S_H R(\bar{Q}_{IO})$, $S_L R(\bar{Q}_{IO})$
	Outcross	$S_L R(\bar{Q}_{IO})$, $S_H R(\bar{Q}_{IO})$	$R(\bar{Q}_{OO})/2$, $R(\bar{Q}_{OO})/2$

where $S_L < 1/2 < S_H$ because $\gamma_I > \gamma_O$ and $\bar{Q}_{II}$ is the amount of quality produced from both players choosing I. A potential Nash equilibrium in this case is (I,I), as inbreeding could be a profitable deviation when the opponent chooses to outcross. To be more exact, the condition for deviating to O when opponent chooses I is:

$$\frac{\alpha_I - \alpha_O}{\beta} > R(\bar{Q}_{II})\frac{1}{2} - R(\bar{Q}_{OI})s_L.$$

Because $\gamma_I > \gamma_O$, $\bar{Q}_{II} > \bar{Q}_{OI}$ and, since $S_L < 1/2$, the right-hand side of the condition must be more than 0. Thus, the only time outcrossing would be profitable is when inbreeding is most costly than outcrossing, $\alpha_I > \alpha_O$, and the quality gain from inbreeding is sufficiently small. Since inbreeding reduces gains over time due to inbreeding depression, more realistically, we would expect the relative gain from inbreeding to decline at lower levels of d (i.e. a non-linear form for the gain function f)

The actions of both players will impact the aggregate diversity stock, $\bar{d}$. If each player inbred every period, we would see $\bar{d} \to 0$, which would lead to genetic improvement stagnating without any recovery. However, if inbreeding is more costly than outcrossing and the gains to inbreeding eventually slow, players would likely not let d drop below a threshold level, call it d^{ME} for "market equilibrium."

Compare this to what we would assume a social planner would choose to maximize the discounted stream of welfare. Imagine that the social planner picks the level of genetic diversity d and that the welfare-maximizing value is d^*. We contend that $d^* > d^{ME}$, meaning that the market equilibrium is a socially inefficient level of genetic diversity. When a firm outcrosses, it benefits the firm and the entire industry by improving the benefits of outcrossing for all firms. The social planner takes into account both of these benefits, leading to a higher stock of genetic diversity in the long run. In contrast, an individual firm only considers its benefit to its own stock, leading to a lower stock of genetic diversity. Assuming a welfare function W, the resulting welfare loss is $W(d^*) - W(d^{ME})$ since firms will use more diversity than is socially optimal to stay ahead of their competitors.

2.2 The Effect of Genomic Testing

Genomic testing allows dairy breeders to determine the genetic traits of dairy bulls far more quickly than before. Before it was introduced, it took five years to determine a dairy bull's genetic traits. Once the bull reached sexual maturity around the age of one year, the bull's semen would be frozen and sent to several farms to produce offspring from the bull. After about 10 months, the offspring would be born and would take roughly another two years to get to production age. After a few years of production, the bull now has production data from its offspring that can be used to calculate its genetic traits, in our model q.

Genomic testing gathers the same information in a far shorter amount of time by taking a DNA sample of the bull and correlating its individual genes to traits. In terms of our model, genomic testing caused an acceleration in the rate of quality gain, that is an increase in γ_I and γ_O. In the case that they both increase the same amount, there would be no change in d^{ME} since the relative benefits of inbreeding have not changed. However, genomic testing disproportionately benefits inbreeding because it makes identifying beneficial genes easier. In particular, it makes identifying which offspring inherited the genes easier and so isolating the beneficial gene is easier than it was before. Since outcrossing is about expanding the number of genes in the firm's genetic families and not isolating a specific gene, outcrossing benefits far less from genomic testing. Moreover, this isolation works best when there are fewer genes to test for in the population, and so genomic testing can more efficiently isolate a population which shares less genes (meaning a population with a lower genetic diversity).

In the model, this would increase γ_I relative to γ_O and lead to a new "genomic" market equilibrium level of genetic diversity, $d^{GE} < d^{ME}$. Since inbreeding is now more valuable, it could be that the new social planner optimum may also be lower, $d^{**} \leq d^*$. Whether genomic testing actually leads to a higher welfare loss depends on i) how much genomic testing decreases d^{GE} and d^{**} and ii) the relationship between W and d. If $d^* - d^{ME} <$

$d^{**} - d^{GE}$, that is, the diversity gap increased, then genomic testing causes a net welfare loss if welfare is improved if $W'(d) > 0$ or diversity increases welfare. If $d^* - d^{ME} = d^{**} - d^{GE}$, the welfare impact of genomic testing depends on whether d is linear or concave in W. If d is linear in W, then $W(d^{**}) - W(d^{GE}) = W(d^*) - W(d^{ME})$, meaning the welfare loss is unchanged. However, if d is concave in W, then $W(d^{**}) - W(d^{GE}) > W(d^*) - W(d^{ME})$, since they are at lower values. Since inbreeding depression would cause the benefits of inbreeding to erode over time, this is most consistent with d being non-linear in W.

The primary prediction of this model is that the introduction of genomic testing increases inbreeding, which potentially exacerbates a welfare loss in the industry. The welfare loss is derived from the fact that breeders excessively inbreed to stay ahead of their competitors, despite the fact that this depletes genetic diversity and leads to less growth in aggregate quality over time. In this paper, our aim is to test one prediction of the model: that genomic testing increases inbreeding. Specifically, we hypothesize that, after genomic testing, the lines that will become the most inbred are lines that were the most popular before genomic testing. Since most bulls would share DNA with these lines, testing their DNA would gain the most information for firms. Taking our model to data, we would then expect inbreeding to increase the most in the popular lines irrespective of quality, as genomic testing will have made inbreeding more beneficial on these lines than before. This would be consistent with the returns to inbreeding increasing relative to outcrossing.

3 Data Description

To understand the role of genomic testing in dairy breeder decision-making, we use data on dairy bull traits and lineages, including the inbreeding rate of each bull. Dairy bull traits in the US are calculated using data on dairy bull offspring producing milk on US dairy farms. The National Dairy Herd Improvement Program (NDHIP) collects, manages, and analyzes data from dairy farms that are members of Dairy Herd Improvement Associations to assess the bull's contribution to the productivity of their offspring (Hutchins and Hueth, 2023). The raw data collected from farms are used to estimate a statistical model that predicts the genetic contribution of an animal as a sire (or dam) after controlling for environmental variables such as herd, age, pedigree, and season effects. The predicted values for each trait from this model are commonly called the **predicted transmitting ability** (PTA) of a trait.

The primary data for our study is from the National Association of Animal Breeders (NAAB), a trade organization representing all of the major livestock genetics companies selling dairy bulls in the US. Dairy bull traits for all bulls sold by members are posted three times a year at the same time that each bull's predicted genetic traits are calculated and posted publicly by the Council on Dairy Cattle Breeding (formerly by the USDA) (Hutchins and Hueth, 2023). Our data comes from the NAAB's published lists from 2001 to 2020 and represents over 15,400 dairy bulls, both foreign and domestic, sold by its members during this period. The Association represents 95% of the livestock genetics companies and cooperatives in the US and Canada[2], making these data broadly representative of what is sold to US dairy farmers.

The NAAB lists also contain information about lineage, which we use to track descendants of a bull. Every animal born in the USA and Canada is assigned a unique ID which includes information about its breed, country of birth. The NAAB data includes these IDs for the bull and both of its parents, allowing us to build a genetic network of relationships from the data. Inbreeding is commonly measured using the **inbreeding coefficient** (F_X) (Wright, 1922), defined as the probability that two genes in an individual were both descended from a single gene in an ancestor.

$$F_X = \sum_{ca=1}^{K} \left(\frac{1}{2}\right)^{n_1+n_2+1} (1 + F_{ca}) \tag{1}$$

[2]`https://www.naab-css.org/history`

Where ca denotes the common ancestor, n_j is the number of generations separating the common ancestor from the sire (dam), and F_{ca} is the inbreeding of the common ancestor. This value can be calculated before the birth of an animal, since it depends on the inbreeding coefficients of its parents. The inbreeding coefficient is reported when a newborn animal is registered in its database.

The genomic-testing program began in 2008, and the first genomically tested bulls had evaluations published in January 2009. García-Ruiz et al. (2016) finds that, after 2010, the generation interval decreased significantly by about four years. Using the list of bulls that are actively sold by members of the National Association of Animal Breeders, we can see in Figure 3 that the new genomically proven bulls, first listed in 2010, have an average age of a little less than two years old when they are first released, versus around five years for daughter-proven bulls.

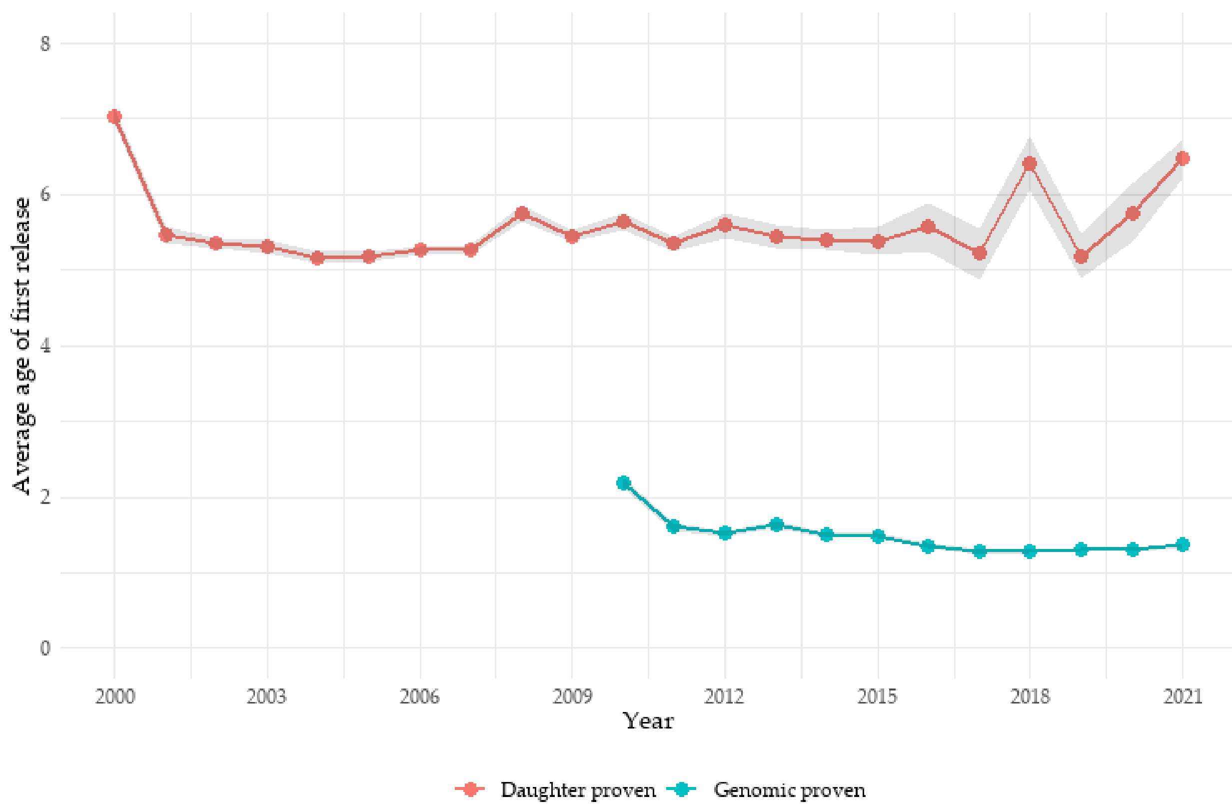

Figure 3: Average age at first proof for Holstein bulls
Source: NAAB

In addition to a significant increase in genetic milk yields, the number of bulls marketed every year has also increased significantly (Hutchins and Funes-Leal, 2025). Figure 4 shows that the number of bulls listed as actively sold in the industry tripled starting in 2010. Such an increment is motivated by the ability to detect bulls with better genetics earlier, thereby allowing companies to sell semen from bulls that have just reached sexual maturity but are not yet at peak performance. As seen in the plot, genomically-tested bulls have become the dominant type of testing in the dairy genetics industry.

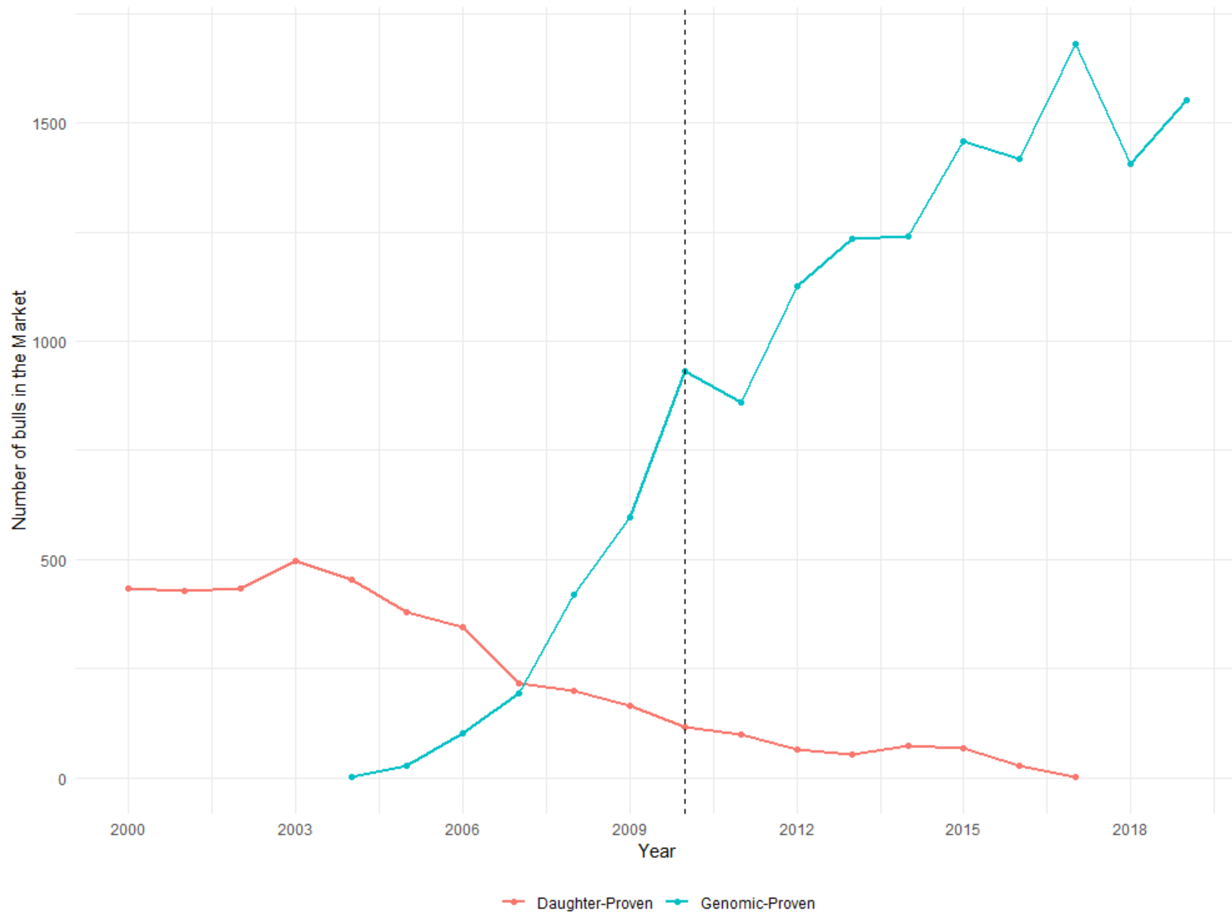

Figure 4: Total number of new bulls released in the market
Source: NAAB

4 Empirical Framework

The goal of this analysis is to determine whether genomic testing increased inbreeding in cattle by increasing the returns to inbreeding among popular genetic lines. Given the high percentage of genomic adoption in the data, it is unlikely we can use ungenotyped bulls as a control group. In an empirical approach, we focus on comparing lines that have similar genetic quality but differ in their pre-existing popularity, measured as number of sons before genomic testing We hypothesize that genomic testing makes inbreeding more valuable on these genetic lines because genomic testing will be more accurate and there are fewer genes to test for. This would suggest that genomic testing has caused breeders to deplete genetic diversity even more than they would have before.

To isolate the role of the existing popularity of genetics, we construct a counterfactual sample with the same genetic traits but from a less popular line. Specifically, we use propensity score matching (Smith and Todd, 2005) to define a control group that resembles the treatment in all observable characteristics, in this case, the genetic traits. If genomic testing led to a race for quality bulls regardless of prior popularity, we should see the inbreeding rate of these lines increase by the same amount as the popular lines. If, instead, genomic testing accelerated line breeding for the popular lines, we should see a significant difference in the inbreeding rates after 2010.

We define a **line** as the set of direct (male) descendants from a particular bull. We also define a "popular" bull as a bull whose number of sons is in the 95th percentile of the distribution of offspring. To construct the line, we start by considering sires born between 1998 and 2004, as the "founders" of each line. Our definition depends critically on when the founders of such lines were born; if we choose too early of a date, lines are too difficult to distinguish since all bulls are related to some degree. After identifying the sires on which to build lines, we recursively build their pedigrees by matching each bull to its sire (father) until we reach the latest animal in the sample. Figure 14 in the appendix summarizes the sample selection mechanism; the treatment and control groups are comprised of the descendants of two groups: popular bulls (treatment) and matched bulls (control).

The first step is to match the founders of such lines with other sires that did not father as many sons but had comparable genetic traits. Second, we find all descendants from both groups and compare the evolution of their inbreeding rates through time. The idea behind this procedure is that when those bulls were in the market, any reasonably similar sire could have been chosen but was not, and thus, can be used to construct a counterfactual line. To do so, we use a propensity score matching estimator, let $p(X) = \Pr(T_j = 1|\mathbf{X})$ where T is a dummy equal to 1 if sire j's offspring is on the 95^{th} percentile of the distribution and X_j be a vector comprised of the following:

1. Predicted Transmitting Ability for production traits (milk, fat, and protein), health traits (mastitis, metritis, productive life), and reproductive traits (daughter pregnancy rate, early first calving, gestation length).

2. Dummies for the absence of certain recessive genetic disorders (Holstein Foundation, 2021) such as *brachyspina*[3], complex vertebral malformation, cholesterol deficiency, and mule foot.

3. Dummies for the presence of genes associated with specific qualitative traits (polled, or red coat color).

4. Dummies for the presence of haplotypes (specific sequences of DNA at different locations in the chromosome) that affect fertility (Holstein Foundation, 2018) when present in homozygous form (likely to be expressed when two carrier animals are mated).

We then estimate the propensity score (Cameron and Trivedi, 2005, p. 873), the probability of being a popular bull given the level of PTAs and the presence of the entire set

[3]A lethal genetic disease that causes spontaneous abortions.

of variables listed above. The validity of our method relies on a series of assumptions (Heckman et al., 1997):

Assumption 1 (Stable Unit Treatment Value Assumption): $(inb^{treatment}, inb^{control}) \perp\!\!\!\perp T/\mathbf{X}$.

The average inbreeding rate of any two lines does not vary with the treatment assigned to the other lines. For each unit, there are no different forms of each treatment level (Imbens and Rubin, 2015), conditional on a set of covariates. This assumption implies that conditional on **X**, the distribution of inbreeding rates for both groups will be the same for all founders, but it must differ for their offspring. Figure 5, shows the extent to which the inbreeding rate distribution of our matched sample resembles that of the treatment group. Unlike unmatched sires, their inbreeding distributions have similar means and variances conditional on the set of genetic traits. In a similar fashion, Figure 13 shows the standardized mean differences of all variables used to estimate the propensity score. Balance is achieved for several variables, notably for all haplotype dummies and several PTAs.

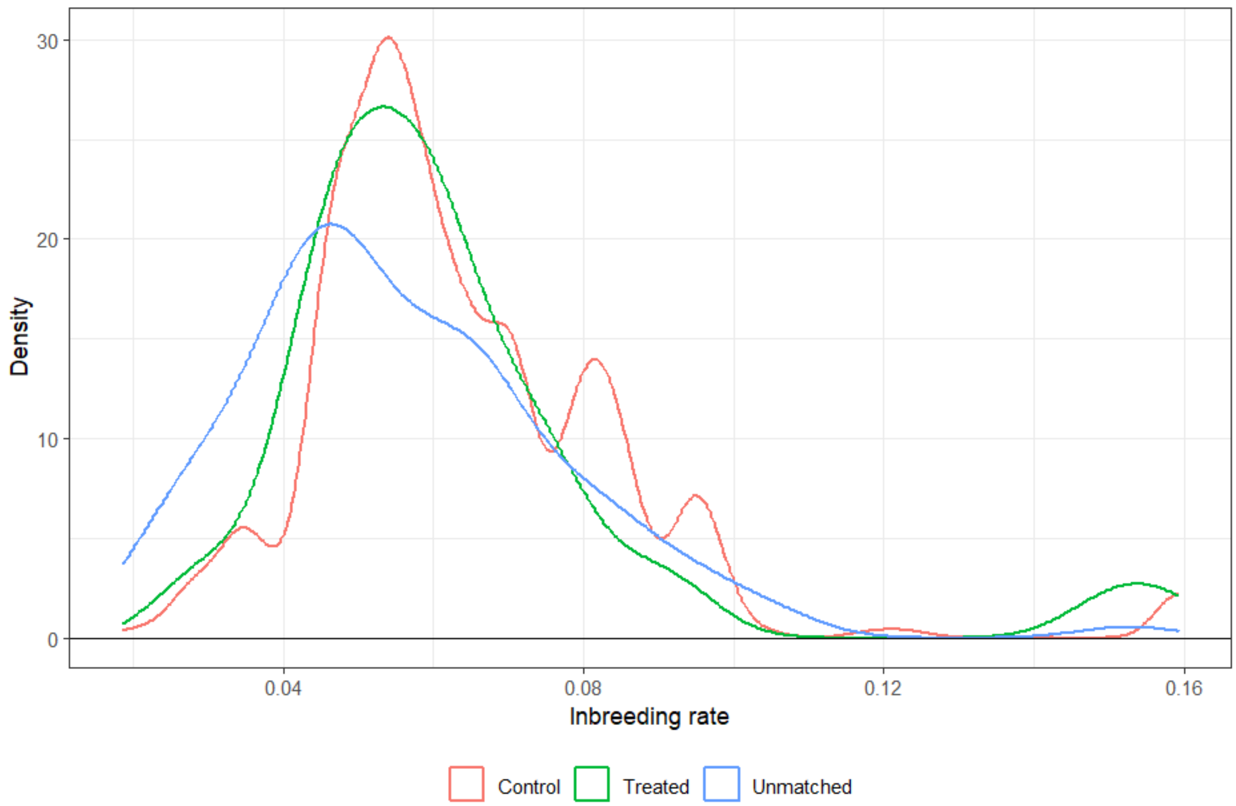

Figure 5: Distribution of Inbreeding rates across pre-treatment groups

Assumption 2: $0 < \Pr\left(T_j = 1 \middle| \mathbf{X}\right) < 1$ for all $\mathbf{X}$

This assumption implies that we can define a propensity score for every value in the treatment group. Assumptions 1 and 2 together are the "strong ignorability" assumption in Rosenbaum and Rubin (1983), and they are necessary in conjunction to identify causal parameters. The set of matching variables above exhausts the entire set of possible selection criteria for breeders; no unobserved variable can influence mating decisions. Figure 6 shows the degree of overlapping between popular bulls and alternative sires; since there is a considerable difference in the size of the groups, popular bulls make up only a tiny fraction of bulls; however, the overlapping is good in the range $0.05 - 0.50$, but there exist a few popular bulls that had few close matches in the sample.

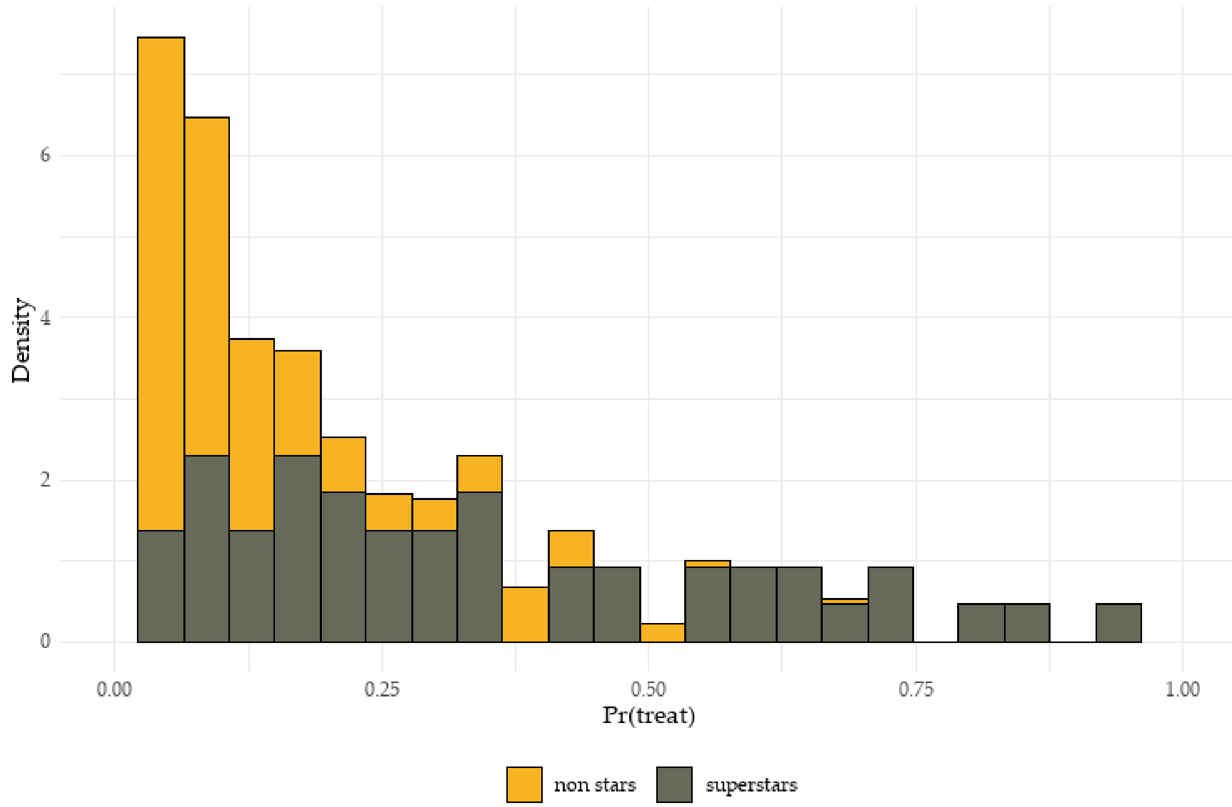

Figure 6: Propensity score by class of bull

A central part of our argument is that treatment and control groups should have similar inbreeding levels before genomic selection. To test that assumption, in Table 1, we calculate the difference in averages of treatment and control groups for inbreeding rates and a few selected traits. There is no statistically significant difference for most traits between treatment and control groups, so our matching method ensures that the difference between both groups is that one bull has been used more often by dairy farmers. Each year, these bulls will have a new set of descendants being born, each with its own inbreeding rate and genetic traits; if our hypothesis holds, the inbreeding rates of the descendants of popular bulls should be significantly higher than those of non-popular descendants.[4]

[4]In Appendix C, we discuss three more assumptions that must hold for our estimator to be valid, including parallel trends, no anticipatory effects, and asymptotic convergence.

Table 1: Balance table for outcome and matched covariates

Variable	Control mean (n = 224)	Treatment mean (n = 43)	t.statistic	p.value	df
Inbreeding rate	2.968	2.904	0.426	0.672	58.147
Milk	0.059	0.059	-0.036	0.971	59.956
Fat	-0.122	-0.170	0.346	0.730	76.548
Protein	-0.851	-1.009	0.416	0.679	61.059
Somatic cell score	-0.151	-0.037	-0.675	0.502	66.764
Productive life	-0.619	-0.677	0.173	0.864	59.829
Daughter preg. rate	-1.224	-0.489	-1.336	0.186	70.101
Heifer conc. rate	-11.953	-5.553	-1.807	0.075*	71.841
Cow conc. rate	-0.420	-0.209	-1.278	0.206	65.480
Livability	-0.393	-0.177	-1.816	0.074*	70.905
Type	-0.827	-0.551	-1.721	0.090*	69.510
Gest. length	0.342	-0.055	1.719	0.090*	64.087
Heifer liv.	-0.569	-0.579	0.032	0.975	58.933
Early first calving	0.101	0.232	-1.549	0.127	58.339
Mastitis	-0.888	-0.764	-0.288	0.774	58.621
Metritis	-0.840	-0.891	0.202	0.840	62.094
Strength	-0.589	-0.581	-0.047	0.963	57.734
Rear legs rear view	-191.265	-24.936	-1.581	0.119	67.713
Rear legs side view	-0.027	-0.013	-0.495	0.623	60.964
Foot leg score	-1.348	-1.123	-0.666	0.508	59.099
Teat rear place	-8.284	-3.255	-1.859	0.067	68.421
Rump angle	-0.360	-0.136	-1.447	0.153	64.334
Thurl width	-0.010	0.050	-0.342	0.734	63.537
Foot angle	-0.767	-0.417	-2.044	0.045**	68.106
Fore udder	-0.743	-0.432	-1.735	0.087*	66.269
Rear udder height	0.142	0.193	-0.254	0.800	60.330
Rear udder width	3.044	2.995	0.690	0.493	52.692
Udder cleft	-0.386	-0.349	-0.235	0.815	77.188
Udder depth	-0.206	-0.267	0.409	0.684	67.108
Teat fron place	-0.343	-0.195	-1.006	0.318	66.497
Teat length	0.197	0.159	0.195	0.846	59.002
Milk fever	-0.312	-0.140	-0.948	0.347	62.811
Stature	-0.255	-0.233	-0.138	0.890	72.664
Dairy form	-0.573	-0.403	-1.442	0.153	73.087
Body depth	-0.401	-0.193	-1.127	0.264	61.777
Sire calv. ease	-0.738	-0.589	-0.989	0.326	73.000
Daughter calv. ease	2.427	2.287	1.188	0.239	63.963

Notes: ***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

We observed the average inbreeding rates of the treatment and matched control groups before and after the introduction of genomic selection. Let $j = 1, \ldots, J$ be the set of lines, defined as the set of direct descendants from a single male ancestor. Each line is comprised of $i = 1, \ldots, n_j$ bulls born in year t and belongs to firm $k = 1, \ldots, K$. If we divide the 2005-2017 period into two groups, pre-intervention (2005-2009) and post-intervention (2010-2019), denoted as 1 and 2 and let $treat_j$ denotes whether a bull from line j belongs to treatment ($treat_j = 1$) or control ($treat_j = 0$) groups, then:

$$inb_{ijkt} = treat_j \times inb_{ijkt}^{treat} + (1 - treat_j) \times inb_{ijkt}^{control}$$

Where inb_{ijkt} is the inbreeding rate of bull i from line j, that was marketed by firm k and was born in year t. To measure the effects of genomic selection on inbreeding, we estimate the differences-in-differences model:

$$inb_{ijkt} = treat_j + (treat_j \times post_t)\beta_{post} + \alpha_t + \alpha_k + \varepsilon_{ijt} \tag{Model 1}$$

$$inb_{ijkt} = treat_j + treat_j \times \sum_{\substack{t=2005 \\ t\neq 2009}}^{2019} \beta_t \mathbf{I}(year = t) + \alpha_t + \alpha_k + \varepsilon_{ijt} \tag{Model 2}$$

$$inb_{ijkt} = treat_j + treat_j \times \sum_{\substack{t=2005 \\ t\neq 2009}}^{2019} \beta_t \mathbf{I}(year = t) + \alpha_t + \alpha_k + \mathbf{X}_{it}\gamma + \varepsilon_{ijt} \tag{Model 3}$$

$$inb_{ijkt} = treat_j + treat_j \times \sum_{\substack{t=2005 \\ t\neq 2009}}^{2019} \beta_t \mathbf{I}(year = t) + \alpha_t + \alpha_k + \mathbf{X}_{it}\gamma + (\mathbf{X}_{it} \times post_t)\,\rho + \varepsilon_{ijt} \tag{Model 4}$$

Model (2) is a specification that does not include any covariates (PTAs) and decomposes the ATT by year of birth cohort. Model (3) incorporates the bull's genetic traits ($\mathbf{X}_{it}$) and firm-level fixed effects, and Model (4) includes interactions of genetic traits and the post-treatment dummy. We include the last specifications to account for trait-level trends. We also know that the treatment affects both inbreeding rates and PTAs, but we do not have a structural model for their impact (Lewbel, 2019), so we include them in a non-linear fashion as both levels and interactions to reduce the uncertainty about the functional form of our controls.

5 Results

Figure 7 shows the treatment and control groups' average inbreeding rates and confidence intervals across years. Genomic selection did not immediately increase inbreeding rates; both curves began to diverge substantially after 2011. Inbreeding rates, in turn, began to increase two years later, and both curves diverged significantly from 2012 onwards.

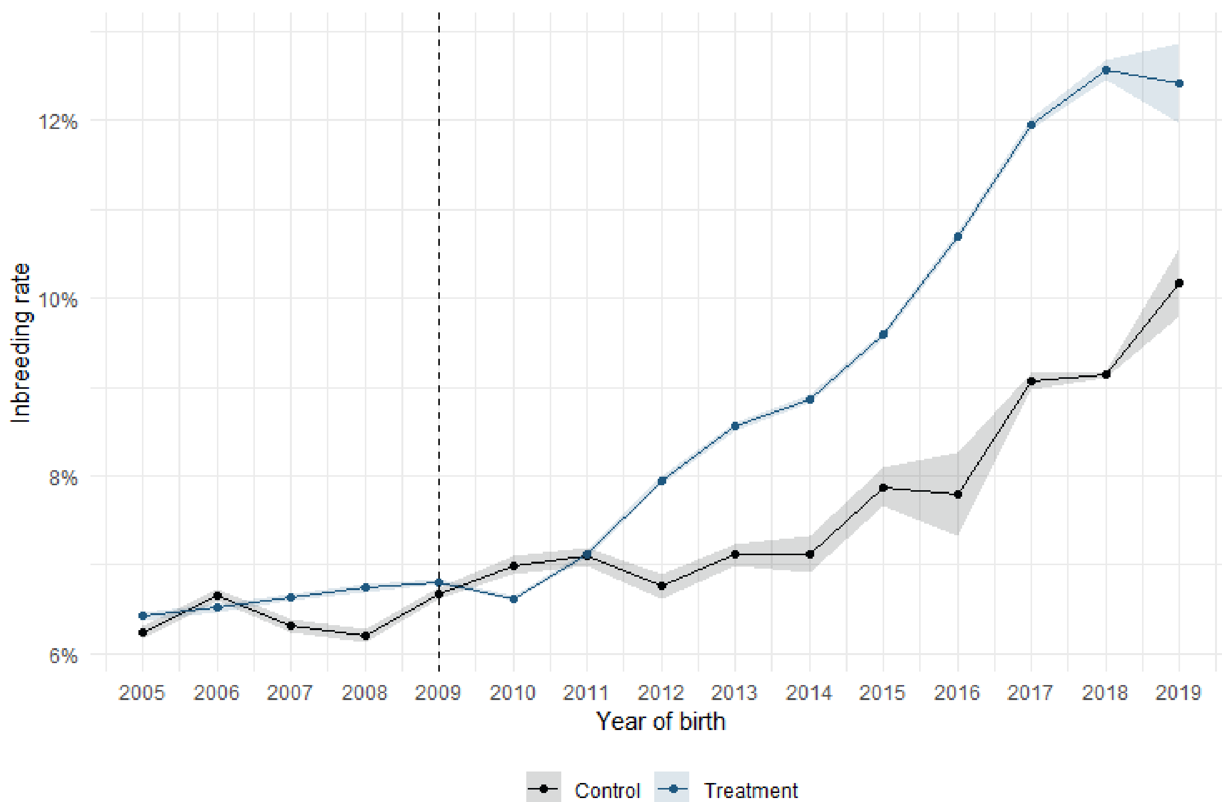

Figure 7: Average inbreeding rates in treatment and control groups

This plot closely matches the evolution of the population-average inbreeding rates shown in Figure 2. This behavior is consistent with our hypothesis that the gains from inbreeding became higher in these lines after genomic testing was introduced. Our findings align with those reported by Guinan et al. (2023), who also examine the temporal trends in productivity and reproductive traits in US dairy cattle. The authors also document a decrease in the generation interval and an increase in inbreeding rates of Holstein and Jersey cattle but do not consider the latter an issue because the expected inbreeding rates of future offspring adjust genetic trait values.

Table 2 shows the results of three models. Model (1) only includes the interaction between treatment and an indicator for the post period, making it a 2x2 difference-in-difference model. Model (2) splits the treatment effect by year of birth averages, Model (3) includes the complete set of genetic traits and firm-level fixed effects, and Model (4) is the model with traits and a post-treatment dummy. In all specifications, standard errors are clustered at the line level to better deal with the inconsistency generated by the autocorrelation of inbreeding rates between different generations (Bertrand et al., 2004).

Model (1) is the simple differences-in-differences estimation where $post_t$ is a dummy equal to 1 if the bull was born after 2009. The results show that bulls descending from popular bulls increased their inbreeding rates by 1.46 percent more than those who descend from less popular bulls, on average. This is about a 48% increase relative to the sample mean. Since the impact of genomic testing on inbreeding rates only became noticeable after 2011, the coefficient underestimates the average treatment effect on the treated (ATT) of genomic selection. Model 2 shows that the treatment-year-of-birth interactions are positive and significant from 2012 onwards, with the 2018 effect being the highest. The popular bulls born in 2018 had an inbreeding increase of 3.5%, meaning they had double the inbreeding rate of the pre-genomics ancestors (2.9%).

Models (3) and (4) in Table 2 show that there is a positive and significant incremental variation for the treated versus control groups in the years 2012-2019 even with the inclusion of controls. In all years after 2010, the coefficients are positive. They both

Table 2: Difference-in-Differences estimates from models 1 to 4

		inbreeding		
		No covariates	Traits	Traits and interactions
treat × post = 1	1.453***			
	(0.4250)			
treat × yob = 2005		0.0944	0.0489	0.0252
		(0.2975)	(0.2988)	(0.2829)
treat × yob = 2006		-0.2183	-0.1182	-0.2789
		(0.3067)	(0.2933)	(0.2534)
treat × yob = 2007		0.2328	0.2826	0.2848
		(0.3038)	(0.2676)	(0.2534)
treat × yob = 2008		0.4035	0.3246	0.3294
		(0.3148)	(0.2798)	(0.2425)
treat × yob = 2010		-0.4589	-0.4721*	-0.5583**
		(0.2827)	(0.2825)	(0.2667)
treat × yob = 2011		-0.0164	-0.0455	-0.1671
		(0.3599)	(0.2596)	(0.2511)
treat × yob = 2012		1.224**	1.252**	1.151*
		(0.4906)	(0.5633)	(0.5981)
treat × yob = 2013		1.433***	1.083**	0.9980**
		(0.4124)	(0.4631)	(0.4725)
treat × yob = 2014		1.780***	1.600***	1.501***
		(0.3017)	(0.3513)	(0.3068)
treat × yob = 2015		1.637***	1.363***	1.255***
		(0.3453)	(0.4159)	(0.4256)
treat × yob = 2016		2.414***	1.766***	1.710***
		(0.2424)	(0.2697)	(0.2564)
treat × yob = 2017		2.849***	2.770***	2.734***
		(0.4456)	(0.4429)	(0.4507)
treat × yob = 2018		3.417***	3.576***	3.515***
		(0.4530)	(0.4566)	(0.4847)
treat × yob = 2019		2.005**	2.071**	2.022**
		(0.8611)	(0.8827)	(0.8785)
p-value for nonzero pre-effect		0.416	0.498	0.107
Number of clusters	185	185	185	185
Observations	15,491	15,491	15,490	15,490
R^2	0.16988	0.47369	0.49938	0.50467
Within R^2	0.00702	0.01235	0.04150	0.05163
treat fixed effects	✓	✓	✓	✓
post fixed effects	✓			
yob fixed effects		✓	✓	✓
parent_company fixed effects			✓	✓

show a positive and increasing inbreeding rate in the treatment group comprised of all descendants from prestige sires relative to the control group of descendants of comparable non-popular sires.[5]

Table 2 also includes the results for the F-test for joint significance of all pre-treatment differences in trends for the interaction of treatment and year of birth. The p-values are not significant at any specific level in any of the specifications.[6]

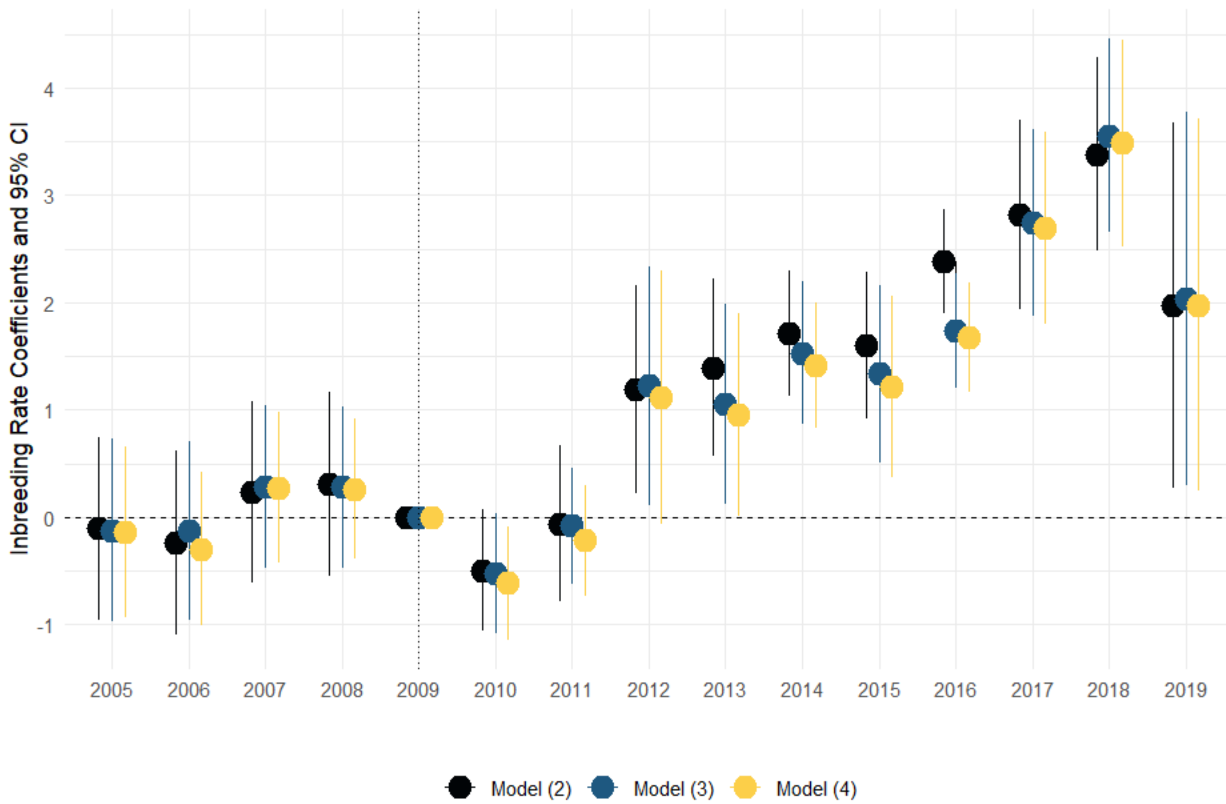

Figure 8: Coefficient plots for different specifications

Figure 8 shows the coefficients from the interaction of year fixed effects and treatment

[5]The last period in the sample is 2019 instead of 2020 because we cannot make inferences after the first due to a reduction in the number of animals in the control sample; the number of comparable descendants is too small for us to consider them a valid comparison group. This drastic reduction is a consequence of lower fertility rates in the control group; since those sires fathered fewer sons, their sons will beget even fewer grandsons, and after a few years, this group collapses.

[6]This is not a test for parallel trends since the assumption relies on unobservable counterfactuals. Rather, it is usually considered evidence in favor of the assumption of parallel trends. However, this is not necessarily the case since this test has low power against a linear or quadratic pre-trend alternative (Roth, 2022). In our case, we can dismiss this concern: since our matching method ensures the founders meet the condition, the entire effect must be due to their sons and grandsons.

status to assess the impact of inbreeding per year. Every year after 2012, the inbreeding rate of the treatment group is between 1 and 3.5 percentage points higher than the control group. The coefficient values decrease slightly after controlling for their genetic traits, but remain statistically significant. The figure also shows that all pre-treatment coefficients are not significant. Figure 8 uses the *sup-t* uniform confidence intervals for the pre-treatment periods of Montiel-Olea and Plagborg-Møller (2019) as suggested by Freyaldenhoven et al. (2019) instead of the standard pointwise confidence. These confidence intervals contain the actual path of the coefficients 95% of the time and can better help understand which types of pre-trends are consistent with the data.

6 Discussion

This article investigates the consequences of genomic selection on the US dairy industry. On the one hand, this new technology significantly improved productivity. On the other hand, we find evidence that it increased the average inbreeding rate of all bulls in the market. As genomic testing increased the rate of genetic improvement, it appeared to exert competitive pressure on animal breeders to select from popular lines with high gains. By not crossing enough from other lines, the inbreeding rate increases and dampens future productivity gains by depleting genetic diversity over the long term. As the average inbreeding rate increases, each new generation of animals faces a higher risk of inbreeding depression and of recessive genetic diseases that may or may not be able to be predicted.

It should be noted, however, that the benefits of genomic testing appear to be much larger than the costs of inbreeding depression in the short term. The USDA publishes a profitability index called Net Merit that estimates the profitability contribution of a given dairy bull based on its genetic traits. Even after correcting for inbreeding depression, we estimate that the index increased by 275 USD per animal relative to what it might have been if genetic traits had not increased after 2010. The largest per-animal cost of inbreeding in our findings is about a 140 USD per animal increase inbreeding cost relative to lines that are not popular. These costs are difficult to compare since one is relative to a control group and the other is not.[7] Regardless, it appears that the net benefits of genomic testing are positive in the short term.

While there are clearly short-term gains to genomic testing, the actual welfare effects are harder to discern. As we explore in our conceptual framework, there may still be welfare losses relative to the counterfactual of no genomic testing. However, our paper does not seek to observe such a counterfactual, instead focusing on the relative increase in inbreeding for popular lines. This is because reducing genetic diversity by inbreeding increases genetic gain in the short run at the cost of more steady growth in productivity in the long run. At its worst, excessive inbreeding leads to a collapse in the population whereby there is no more genetic diversity to continue growing production.

Because dairy cattle breeders are not incentivized to cross-breed in the current market, this suggests a role for the government or an industry group to help address the coordination problem. Currently, the Holstein Association of the United States adjusts the values of genetic traits based on inbreeding levels to penalize breeders who select bulls with high inbreeding rates, reflecting the higher reproductive costs of such animals. Yet, Cole (2024) points out that many of the most inbred bulls have such high levels of traits that the downgrading will not temper the demand for them enough. Another option would be to give dairy farmers more information about the dangers of inbreeding to reduce demand for the most inbred lines (Cole, 2024).

More research is needed in this area in agricultural economics, as the coordination problems of dairy cattle breeding have not yet been examined by the field. Future work could make a significant contribution by expanding our conceptual framework to understand what the socially optimal level of inbreeding is and under what circumstances does a technology like genomic testing actually deepen the welfare losses from inbreeding. A

[7] See Appendices A and B for more details on our calculations.

full cost-benefit analysis of genomic testing is, unfortunately, outside the scope of this paper. Yet, studying the increase in inbreeding is crucial for ensuring a safe and global resilient dairy sector in the future and ensuring that US dairy cattle breeders continue to sustainably raise the productivity of dairy cattle in the future.

References

Adams, H. A., Sonstegard, T. S., VanRaden, P. M., Null, D. J., Van Tassell, C. P., Larkin, D. M., and Lewin, H. A. (2016). Identification of a nonsense mutation in APAF1 that is likely causal for a decrease in reproductive efficiency in holstein dairy cattle. *Journal of Dairy Science*, 99(8):6693–6701.

Bertrand, M., Duflo, E., and Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *The Quarterly Journal of Economics*, 119(1):249–275.

Brown, Z. S. (2018). Voluntary programs to encourage refuges for pesticide resistance management: lessons from a quasi-experiment. *American Journal of Agricultural Economics*, 100(3):844–867.

Cameron, A. C. and Trivedi, P. (2005). *Microeconometrics: Methods and Applications*. Cambridge University Press, Cambridge, UK.

Cole, J. (2019). The truth about inbreeding in dairy cattle. *Hoard's Dairyman*.

Cole, J. and VanRaden, P. (2011). Use of haplotypes to estimate mendelian sampling effects and selection limits. *Journal of Animal Breeding and Genetics*, 128(6):446–455.

Cole, J. B. (2024). Perspective: Can we actually do anything about inbreeding? *Journal of Dairy Science*, 107(2):643–648. Publisher: Elsevier.

Council on Dairy Cattle Breeding (2025). Impact on u.s. dairy. https://uscdcb.com/impact/.

Daetwyler, H., Villanueva, B., Bijma, P., and Woolliams, J. (2007). Inbreeding in genome-wide selection. *Journal of Animal Breeding and Genetics*, 124(6):369–376.

Epanchin-Niell, R. S. and Wilen, J. E. (2015). Individual and cooperative management of invasive species in human-mediated landscapes. *American Journal of Agricultural Economics*, 97(1):180–198.

Freyaldenhoven, S., Hansen, C., and Shapiro, J. M. (2019). Pre-event trends in the panel event-study design. *American Economic Review*, 109(9):3307–38.

García-Ruiz, A., Cole, J. B., VanRaden, P. M., Wiggans, G. R., Ruiz-López, F. J., and Tassell, C. P. V. (2016). Changes in genetic selection differentials and generation intervals in us holstein dairy cattle as a result of genomic selection. *Proceedings of the National Academy of Sciences*, 113(28):E3995–E4004.

Guinan, F., Wiggans, G., Norman, H., Dürr, J., Cole, J., Van Tassell, C., Misztal, I., and Lourenco, D. (2023). Changes in genetic trends in us dairy cattle since the implementation of genomic selection. *Journal of Dairy Science*, 106(2):1110–1129.

Heckman, J. J., Ichimura, H., and Todd, P. E. (1997). Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *The Review of Economic Studies*, 64(4):605–654.

Holstein Foundation (2018). Interpreting and Utilizing Haplotype Information. `https://www.holsteinusa.com/pdf/Interpreting_and_Utilizing_Haplotype_Information_1218.pdf`. Accessed: 2023-14-06.

Holstein Foundation (2021). Understanding Genetics and the Sire Summaries. `https://forms.gle/bfB57tLEijiorH7u8`. Accessed: 2023-14-06.

Hutchins, J. and Funes-Leal, V. (2025). After a decade of genomic testing, are milk yields being impacted? *Choices: The Magazine of Food, Farm, and Resource Issues*, 40(4).

Hutchins, J. and Hueth, B. (2023). 100 years of data sovereignty: Cooperative data governance and innovation in us dairy. *Applied Economic Perspectives and Policy*, pages 1–26.

Imbens, G. W. and Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press.

Kerr, W. A. (1984). Selective breeding, heritable characteristics and genetic-based technological change in the canadian beef cattle industry. *Western Journal of Agricultural Economics*, 9(1):14–28.

Ladd, G. W. and Gibson, C. (1978). Microeconomics of technical change: What's a better animal worth? *American Journal of Agricultural Economics*, 60(2):236–240.

Lence, S. H. and Singerman, A. (2023). When does voluntary coordination work? evidence from area-wide pest management. *American Journal of Agricultural Economics*, 105(1):243–264.

Lewbel, A. (2019). The identification zoo: Meanings of identification in econometrics. *Journal of Economic Literature*, 57(4):835–903.

Livingston, M. J., Carlson, G. A., and Fackler, P. L. (2004). Managing resistance evolution in two pests to two toxins with refugia. *American Journal of Agricultural Economics*, pages 1–13.

Lozada-Soto, E. A., Maltecca, C., Jiang, J., Cole, J. B., VanRaden, P. M., and Tiezzi, F. (2024). Effects of germplasm exchange strategies on genetic gain, homozygosity, and genetic diversity in dairy stud populations: A simulation study. *Journal of Dairy Science*, 107(12):11149–11163.

Makanjuola, B. O., Miglior, F., Abdalla, E. A., Maltecca, C., Schenkel, F. S., and Baes, C. F. (2020). Effect of genomic selection on rate of inbreeding and coancestry and effective population size of holstein and jersey cattle populations. *Journal of Dairy Science*, 103(6):5183 – 5199.

Melton, B. E., Colette, W. A., and Willham, R. L. (1994). Imputing input characteristic values from optimal commercial breed or variety choice decisions. *American Journal of Agricultural Economics*, 76(3):478–491.

Melton, B. E., Heady, E. O., and Willham, R. L. (1979). Estimation of economic values for selection indices. *Animal Production*, 28(3):279–286.

Montiel-Olea, J. L. and Plagborg-Møller, M. (2019). Simultaneous confidence bands: Theory, implementation, and an application to SVARs. *Journal of Applied Econometrics*, 34(1):1–17.

Richards, T. J. and Jeffrey, S. R. (1996). Establishing Indices of Genetic Merit Using Hedonic Pricing: An Application to Dairy Bulls in Alberta. *Canadian Journal of Agricultural Economics/Revue canadienne d'agroeconomie*, 44(3):251–264.

Rosenbaum, P. R. and Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55.

Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–22.

Roth, J., Sant'Anna, P. H., Bilinski, A., and Poe, J. (2023). What's trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218 – 2244.

Schroeder, T. C., Espinosa, J. A., and Goodwin, B. K. (1992). The value of genetic traits in purebred dairy bull services. *Review of Agricultural Economics*, 14(2):215–226.

Smith, J. A. and Todd, P. E. (2005). Does matching overcome Lalonde's critique of non-experimental estimators? *Journal of Econometrics*, 125(1):305–353. Experimental and non-experimental evaluation of economic policy and models.

Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175 – 199.

Van Raden, P., Neupane, M., Toghiani, S., Parker Gaddis, K., and Tempelman, R. (2021).

Net merit as a measure of lifetime profit: 2021 revision. Technical report, USDA-AIP Research Report.

Wechsler, S. and Smith, D. (2018). Has resistance taken root in us corn fields? demand for insect control. *American Journal of Agricultural Economics*, 100(4):1136–1150.

Wright, S. (1922). Coefficients of inbreeding and relationship. *The American Naturalist*, 56(645):330–338.

Wuepper, D., Henzmann, S., and Finger, R. (2023). Measuring farmer time preferences: A systematic literature review for europe and north america. *Journal of the Agricultural and Applied Economics Association*, 2(4):823–837.

A Genomic Selection Cost Appraisal

Using published estimates of the costs of inbreeding, we can assign a monetary value to increases in inbreeding rates. The Council of Dairy Cattle Breeding (CDCB) estimates that each additional percentage of inbreeding is associated with a 40.11 USD reduction in income from dairy product sales. The total cost depends on the average inbreeding rate we consider.

We can use the coefficients from Model (4), our preferred specification, as the conditional difference in inbreeding rates between the control and treatment groups. In other words, we can assign monetary values to the Average Treatment on the Treated values (ATT) in our sample. We cannot extrapolate this result to the entire set of bulls, since we did not choose our sample at random. The per-animal average increased cost in year t is equal to:

$$\overline{Cost}_t = 40.11 \times \hat{\beta}_t \tag{2}$$

Next, we multiply this value by the total number of Holstein cattle in the US after the introduction of Genomic Selection (2012-2019). Finally, we can calculate their net present value using a discount rate (r) equal to 4.5%:

$$\begin{aligned} NPV &= \sum_{t=2012}^{2019} \frac{1}{(1+r)^{t-2012}} N_t \times \overline{Cost}_t \\ &= \sum_{t=2012}^{2019} \frac{1}{(1+r)^{t-2012}} N_t \times \left(40.11 \times \hat{\beta}_t\right) \end{aligned}$$

We then calculate the NPV in 2012 for both cost estimation methods, and they amount to a total USD 4.2 billion for the American dairy industry.

The literature review by Wuepper et al. (2023) shows that farmers in North America and Western Europe have higher discount rates than the market rate, around 23% per annum. Such numbers significantly reduce the NPV of costs to USD 1.6. This value, however, must be read cautiously, since it reflects the costs for one side of the market and represents an average of a very heterogeneous group of firms.

B Benefits from Genomic Selection

While genomic testing increased inbreeding, it also increased milk production and thus profits for the dairy industry.

The benefits of introducing genomic selection in our sample can be calculated using an index called **Net Merit**, which measures the net profit over the lifetime of the average daughter of the animal, expressed in U.S. dollars. Net merit is calculated by multiplying the values of genetic traits by a set of dollar values intended to capture the "market value" of all traits.[8]

Published values of Net Merit are already adjusted for inbreeding, the quality increase will be higher than the losses from inbreeding depression. A more accurate estimation of the benefits requires additional information than the one we have; we need information on the expected future inbreeding for all animals in our sample. We do not have the information required to perform such a calculation.[9]

We can approximate the benefits of genomic selection by looking at the average net merit across the treatment and control groups, in a manner similar to the regression coefficients from the previous section.

Let $\overline{NM}_t^{treatment}$, $\overline{NM}_t^{control}$ be the average net merit index for treatment and control groups in year t, then the NPV of the difference between both groups will be:

[8]Table 4 shows what are the magnitudes of such weights (Van Raden et al., 2021). Note that the calculation involves the usage of subindexes for health and reproduction traits, as well as an estimation of residual feed intake per animal.

[9]Let g_k be the reported PTA of trait k from bull i and $\tilde{g}_{ik}$ be the unadjusted PTA, then their relationship will be: $\tilde{g}_{ik} = g_{ik} - \left[(EFI_i - F_i)\hat{\beta}_k\right]$, where EF_i is the expected future inbreeding and $\hat{\beta}_k$ is the regression coefficient from a linear model that relates observed values with phenotypic values.

$$
\begin{aligned}
NPV_{NM} &= \sum_{t=2012}^{2019} \frac{1}{(1+r)^{t-2012}} N_t \times \Delta NM_t \\
&= \sum_{t=2012}^{2019} \frac{1}{(1+r)^{t-2012}} N_t \times \left(\overline{NM}_t^{treatment} - \overline{NM}_t^{control} \right)
\end{aligned}
$$

Using an interest rate of 4.5%, the NPV of benefits amounts to 6 billion USD in 2012. On the other hand, if we use an interest rate of 23%, the NPV decreases to 2.7 billion USD for the same year, but the same caveats mentioned in the cost estimation also hold for benefits.

C Additional Estimator Assumptions

Our causal estimand of interest is the Average Treatment on the Treated (ATT) in the post-intervention period:

$$
ATT_{post} = E\left[inb_{post}^{treat} - inb_{post}^{control} \middle| T_j = 1 \right]
$$

This quantity measures the average causal effect on the treated lines in the post-treatment period (Roth et al., 2023). To correctly identify causal effects, a series of assumptions must hold (Sun and Abraham, 2021):

Assumption 3 (Parallel trends):

$$
E\left[inb_{post}^{control} - inb_{pre}^{control} \middle| T_j = 1 \right] = E\left[inb_{post}^{control} - inb_{pre}^{control} \middle| T_j = 0 \right]
$$

We assume that the average inbreeding rates for both groups would have evolved in parallel without introducing genomic selection. This assumption is not testable because we cannot observe untreated inbreeding rates after 2009, but it constrains our model specification to a Two-Way Fixed Effects specification we call Model (1):

$$inb_{ikt} = treat_i + (treat_i * post_t)\beta + post_t + \alpha_k + \varepsilon_{ijt} \qquad \text{(Model 1)}$$

Where inb_{ikt} is the inbreeding rate of bull i born from a line that belongs to either the treatment or control group born in year t, supplied by firm k, β_t is a year of birth fixed effect, and α_j is a firm fixed effect.

Assumption 4 (No anticipatory effects): $inb_{pre}^{control} = inb_{pre}^{treat}$

This assumption ensures that both groups were comparable before the treatment; inbreeding rates of popular lines did not increase before the introduction of genomic selection; we use matching methods to ensure that this condition holds.

Under Assumptions 3 and 4, we have that:

$$ATT_{post} = E\left[inb_{post}^{treat} - inb_{post}^{control} \middle| T_j = 1\right] - E\left[inb_{pre}^{treat} - inb_{pre}^{control} \middle| T_j = 0\right] \qquad (3)$$

The ATT is identified as the difference-in-differences of observed outcomes in the pre and post-intervention periods. All elements in this equation are observed and, therefore, estimable. Suppose popular bulls had superior traits to our matched control group. In that case, this condition is violated, and we would not be able to parse out the causal effect of inbreeding from the effect of the bull's genetic superiority. Assumptions 3 and 4 ensure that the OLS estimates $\hat{\beta}$ from Model (2) are consistent and asymptotically valid (Roth et al., 2023).

Assumption 5: Let $W_j = (inb_{j,pre}, inb_{j,post}, T_j)'$ be a vector of outcomes and treatment status for unit i. Then, we observe a sample of N independent and identically distributed random draws $W_i \sim F_W(w)$ for a distribution F satisfying parallel trends.

Under Assumptions 3, 4, and 5: $\sqrt{N}(\hat{\beta} - ATT_{post}) \xrightarrow{D} N(0, \sigma^2)$, then we can consistently estimate the variance σ^2 clustering at the line level (Bertrand et al., 2004) asymptotically ($N \to \infty$ and T fixed).

We can also decompose the variations in treatment time since every year, a certain number of bulls will be born from lines in both groups, each with its inbreeding rate that can be averaged across treatment and control groups. Models 2-4 show the complete specification with leads and lags of the treated group. These models decompose the estimated ATT relative to the treatment average by birth year. Sun and Abraham (2021) call this specification "Dynamic specification." However, there is no staggered treatment since genomic selection affects all lines in the same year, but inbreeding rates evolve independently after the treatment.

This assumption states that the treatment status of an individual from any given line is not affected by the treatment status of others. In our setting, the treatment (genomic selection) affects all individuals, so there is no selection into treatment.

D Robustness Checks

D.1 Total number of descendants

In our analysis, we defined a "popular bull" as a bull whose total number of descendants was among the 95^{th} percentile of their distribution by 2021. The cutoff point also influences this calculation since genomic testing leads to an expansion of the supply of bulls, and the size of any line is endogenous to genomic selection. It can be argued that selecting bulls on the ex-post number of sons is tautological to our argument.

Figure 9 plots the coefficients for the different differences-in-differences specifications, but this time, the number of descendants is calculated only up to 2010, so we assume that genomic selection is just being introduced, and we are trying to guess which lines will be more successful. Coefficients for 2014 and 2015 are now substantially reduced and become insignificant, but the overall result remains unchanged; we still have that the treatment group has notoriously higher inbreeding rates than the control group, as seen in the 2×2 model, where the coefficient is still positive and significant for the entire period, as seen on 5.

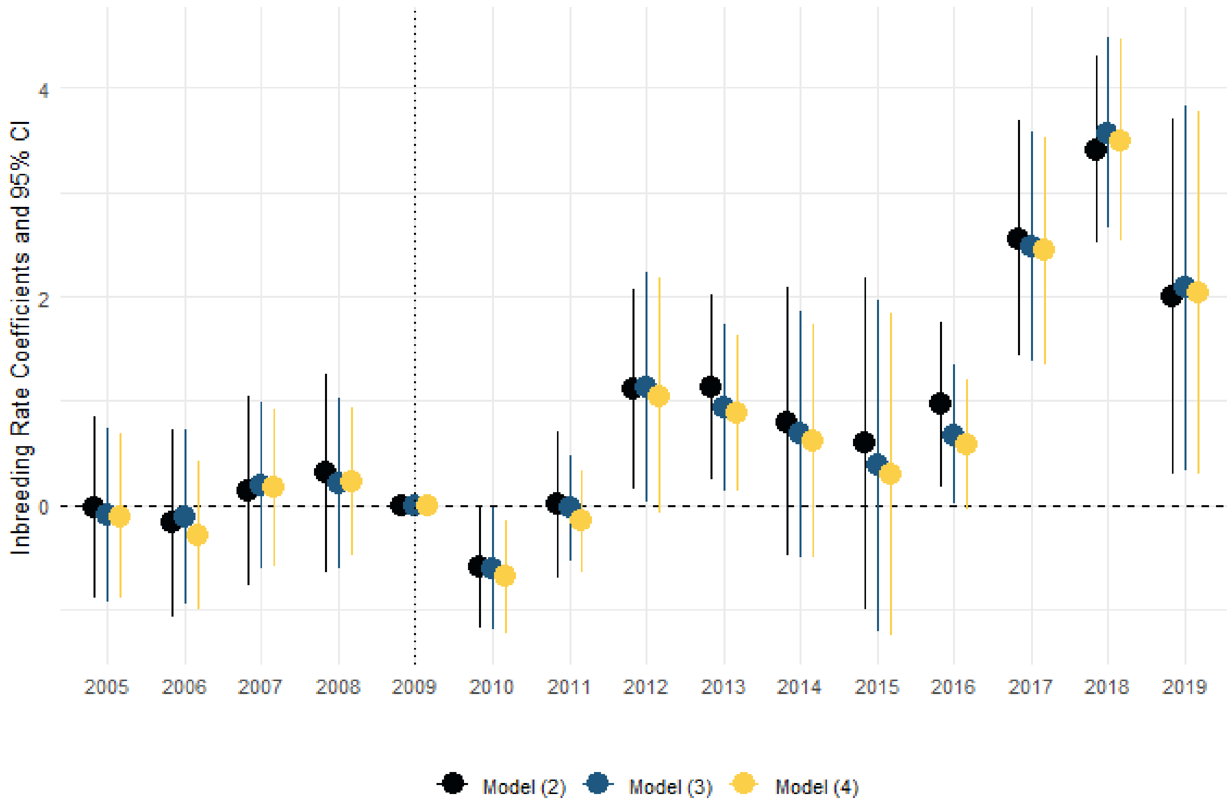

Figure 9: Coefficient plots for an alternative definition of popular bull

D.2 Quarter of birth effects

Since we have the exact date of birth of every animal in the sample, we can group them by a quarter of birth as well; Figure 10 shows the results from Models 2-4 but replacing year for a quarter of birth interactions with the treatment. The signs and magnitudes of coefficients do not differ substantially from the original formulation. The increase in coefficient values is monotonic and statistically significant since 2012, but having a smaller sample size per quarter makes confidence intervals wider.

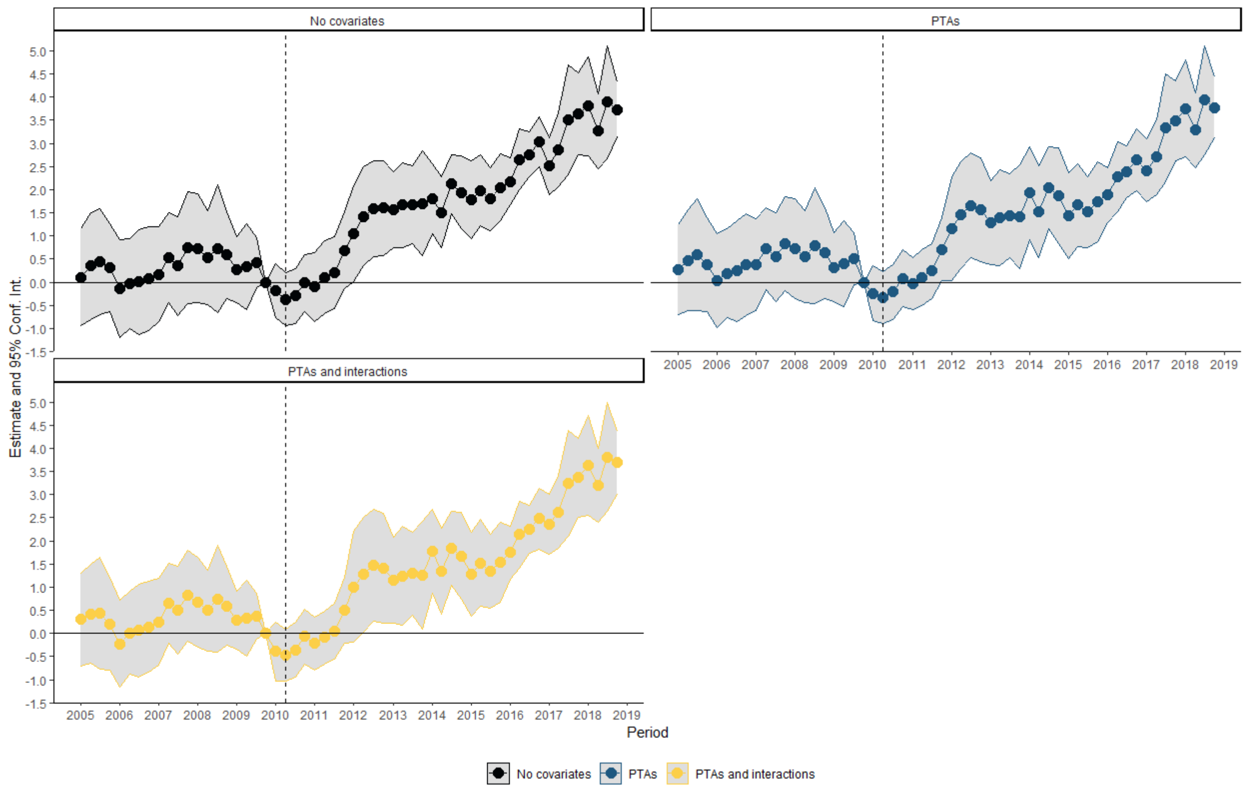

Figure 10: Coefficient plots for quarter of birth effects

This article's results depend heavily on a somewhat arbitrary definition of a "popular" bull in terms of the size of his line. What would happen if we changed that from the 95th to another quantile of the line size distribution? Figure 11 shows the coefficient plots for different definitions of "popular".

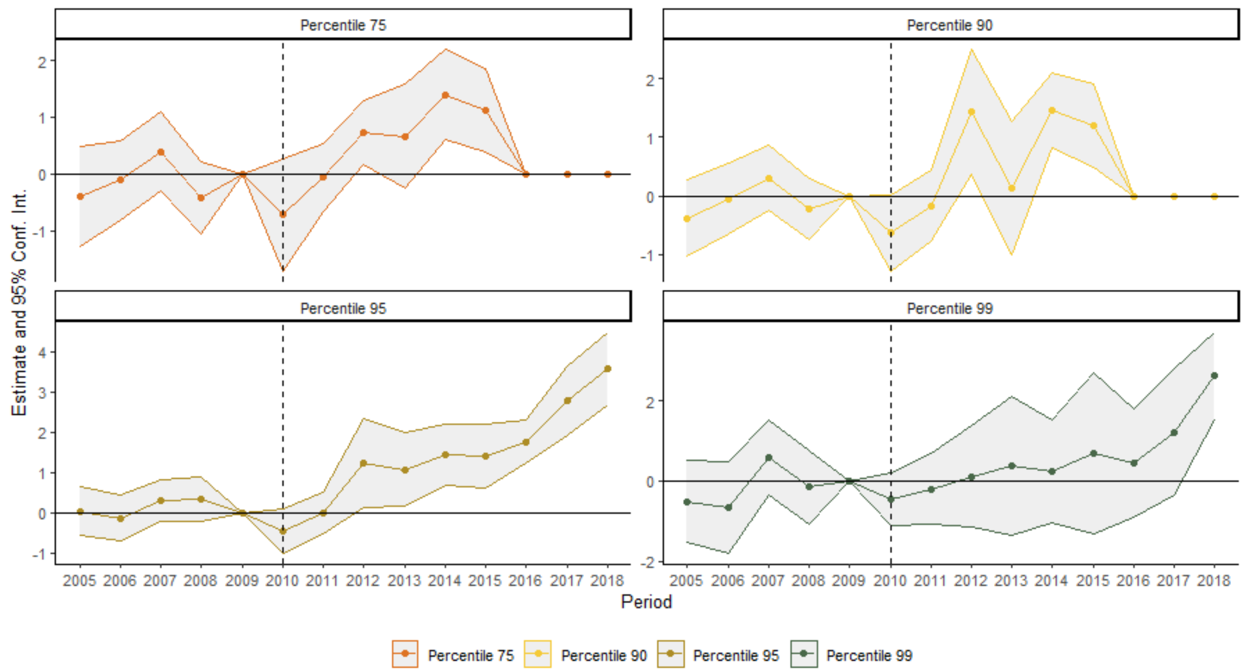

Figure 11: Robustness test: changes in the definition of "popular"

If we change the cutoff point from the 95th to the 99th percentile, the sign, and magnitude of our coefficients do not change much. Still, the confidence intervals widen significantly because our treatment group comprises such specific animals that the matching algorithm cannot find suitable matches. As a consequence, the significance of the difference-in-differences coefficients decreases significantly. Going in the other direction, we can find marginally better matches for the popular group. However, the control group ended abruptly in 2015 because most of the original group's control lines are now included in the treatment, making the control group short-lived. We conclude that our results depend critically on a set of lines that allow for a good comparison when included in the control group; otherwise, the analysis yields insignificant results.

D.3 Date of birth of popular bulls

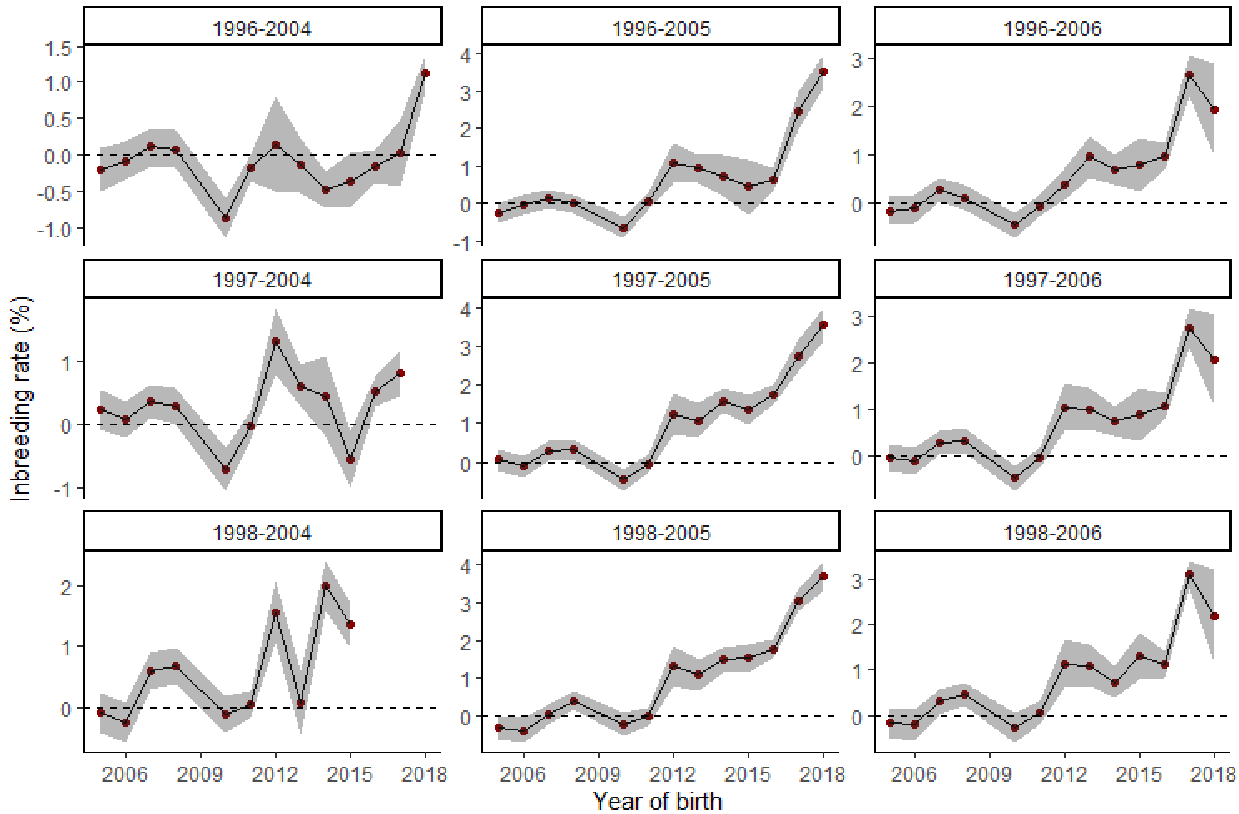

Figure 12: Coefficients from model (2) per different definitions of founders' date of birth

Finally, Figure 12 shows coefficient plots by different definitions of the founders' date of birth. We defined founders as all bulls born between 1997 and 2005, but such a definition is also arbitrary. If we adopt different cutoffs for the period where the founders were born from 1996/1998 to 2004/2006, we have $3 \times 3 = 9$ distinct combinations for starting and end date. The coefficient plot shows that the bulk of the effect can be attributed to sires born from 1996 to 2006; if we extend further the selection period of sires, we will have more endogeneity issues since, by this date, it was known that the bovine genotype was being sequenced, and thus, that there would be a significant effect on the market.

Table 3: Difference-in-Differences estimates from models 1 to 4, using two-way clustering

	inbreeding			
	No covariates		Traits	Traits and interactions
	(1)	(2)	(3)	(4)
treat × post = 1	1.453***			
	(0.3835)			
treat × yob = 2005		0.0944	0.0489	0.0252
		(0.2673)	(0.3287)	(0.2863)
treat × yob = 2006		-0.2183	-0.1182	-0.2789
		(0.2605)	(0.2418)	(0.2168)
treat × yob = 2007		0.2328	0.2826	0.2848
		(0.2653)	(0.2268)	(0.2325)
treat × yob = 2008		0.4035	0.3246	0.3294
		(0.2766)	(0.2691)	(0.2333)
treat × yob = 2010		-0.4589	-0.4721*	-0.5583**
		(0.2711)	(0.2553)	(0.2404)
treat × yob = 2011		-0.0164	-0.0455	-0.1671
		(0.3913)	(0.3461)	(0.3430)
treat × yob = 2012		1.224**	1.252**	1.151*
		(0.4998)	(0.5625)	(0.5882)
treat × yob = 2013		1.433***	1.083**	0.9980*
		(0.4299)	(0.4956)	(0.5077)
treat × yob = 2014		1.780***	1.600***	1.501***
		(0.3256)	(0.3654)	(0.3079)
treat × yob = 2015		1.637***	1.363***	1.255**
		(0.4425)	(0.4464)	(0.4482)
treat × yob = 2016		2.414***	1.766**	1.710**
		(0.4178)	(0.6123)	(0.6431)
treat × yob = 2017		2.849***	2.770***	2.734***
		(0.4665)	(0.4543)	(0.4625)
treat × yob = 2018		3.417***	3.576***	3.515***
		(0.4166)	(0.4422)	(0.4841)
treat × yob = 2019		2.005**	2.071**	2.022**
		(0.8292)	(0.8745)	(0.8532)
p-value for nonzero pre-effect		0.416	0.498	0.107
Number of clusters	257	257	257	257
Observations	15,491	15,491	15,490	15,490
R^2	0.16988	0.47369	0.49938	0.50467
Within R^2	0.00702	0.01235	0.04150	0.05163
treat fixed effects	✓	✓	✓	✓
post fixed effects	✓			
yob fixed effects		✓	✓	✓
parent_company fixed effects			✓	✓

E Additional Plots

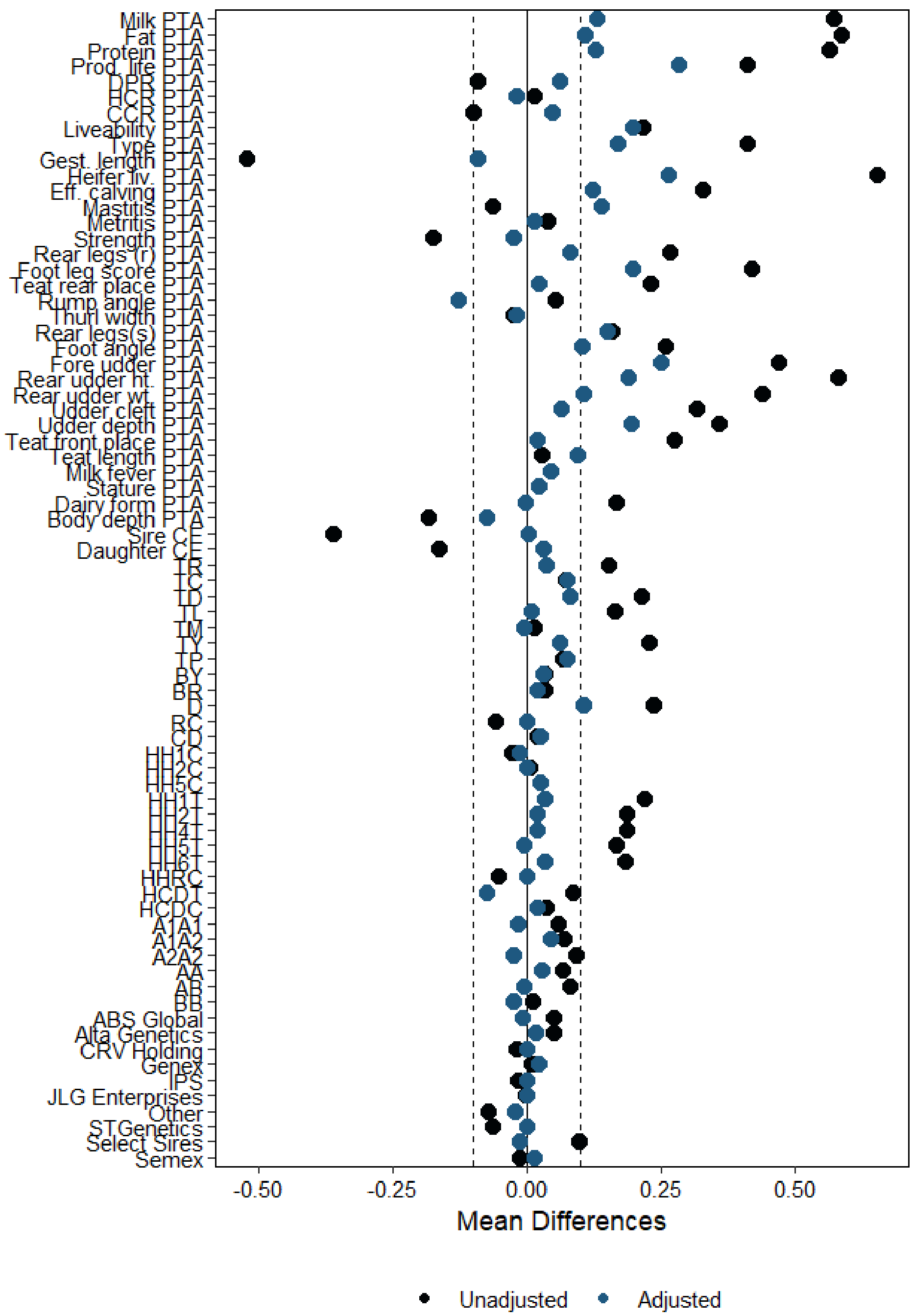

Figure 13: Covariate balance plot

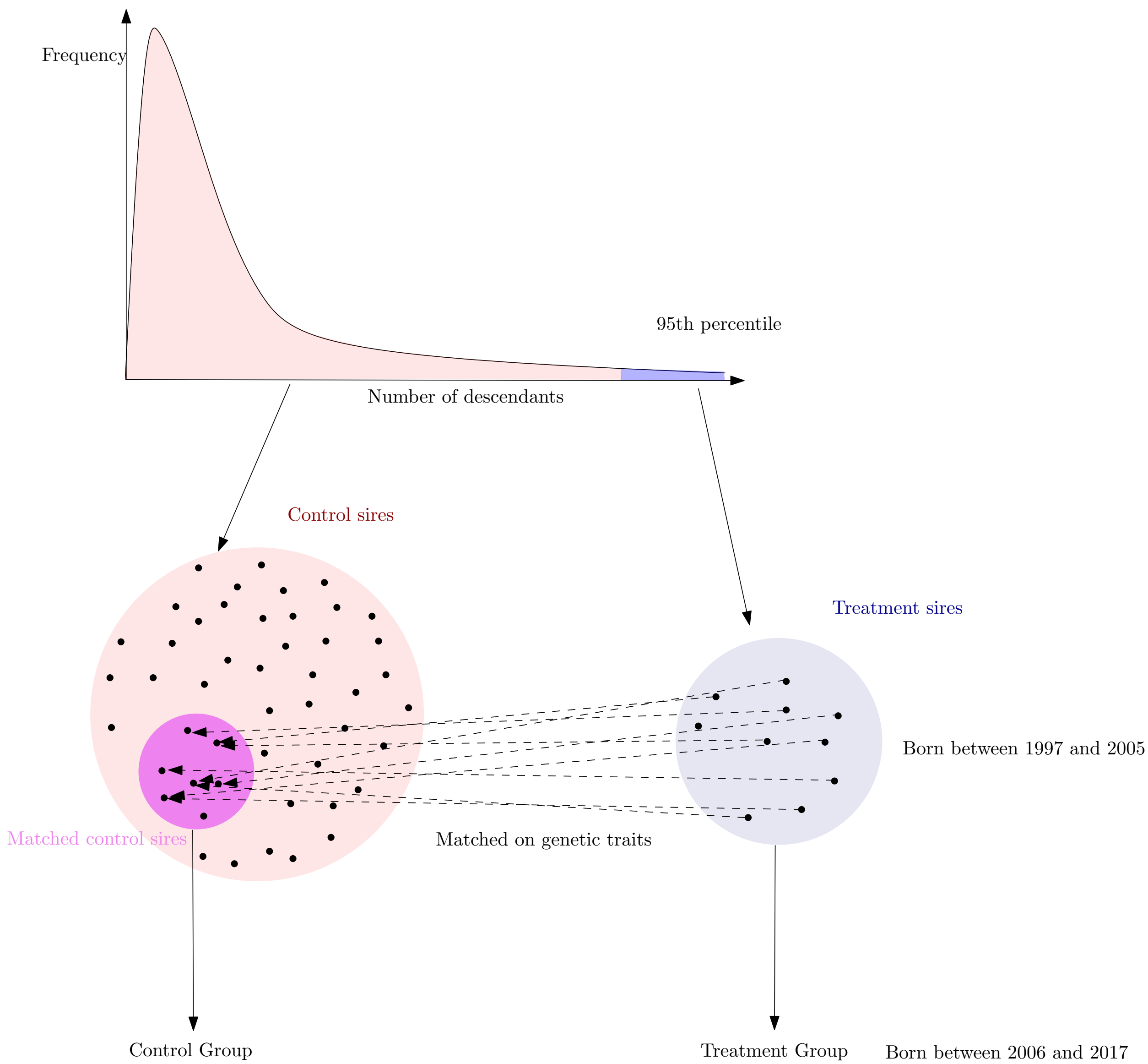

Figure 14: Sample selection mechanism

Trait	Units	Value ($/PTA unit)
Milk	Pounds	0.002
Fat	Pounds	4.18
Protein	Pounds	4.67
Productive Life	Months	34
Somatic Cell Score	Log	-74
Body Weight Composite	Composite	-45
Udder Composite	Composite	19
Foot/Leg Composite	Composite	3
Daughter Pregnancy Rate	Percent	11
Calving Trait Subindex	USD	1
Heifer Conception Rate	Percent	1.1
Cow Conception Rate	Percent	2.2
Livability	Percent	9.8
Health Trait Subindex	USD	1.0
Residual Feed Intake	Pounds	46.2
Effective Calving	Days	2.1
heifer Livability	percent	5.0

Table 4: Economic values for **Net Merit**

Table 5: Difference-in-Differences estimates from alternative scenario

			inbreeding	
	No covariates		Traits	Traits and interactions
	(1)	(2)	(3)	(4)
treat × post = 1	1.217***			
	(0.4671)			
treat × yob = 2005		-0.0189	-0.0863	-0.1019
		(0.2991)	(0.2966)	(0.2842)
treat × yob = 2006		-0.1702	-0.1060	-0.2813
		(0.3088)	(0.2951)	(0.2520)
treat × yob = 2007		0.1376	0.1949	0.1698
		(0.3131)	(0.2826)	(0.2699)
treat × yob = 2008		0.3104	0.2154	0.2295
		(0.3286)	(0.2896)	(0.2508)
treat × yob = 2010		-0.5930**	-0.6049**	-0.6837**
		(0.2916)	(0.2935)	(0.2705)
treat × yob = 2011		0.0069	-0.0263	-0.1519
		(0.3567)	(0.2577)	(0.2509)
treat × yob = 2012		1.115**	1.134**	1.052*
		(0.4877)	(0.5545)	(0.5685)
treat × yob = 2013		1.131**	0.9319**	0.8830**
		(0.4479)	(0.4026)	(0.3758)
treat × yob = 2014		0.8006	0.6818	0.6248
		(0.6513)	(0.6000)	(0.5656)
treat × yob = 2015		0.5968	0.3803	0.3005
		(0.8029)	(0.8051)	(0.7810)
treat × yob = 2016		0.9684**	0.6753**	0.5804*
		(0.3983)	(0.3389)	(0.3175)
treat × yob = 2017		2.558***	2.484***	2.437***
		(0.5683)	(0.5576)	(0.5518)
treat × yob = 2018		3.407***	3.567***	3.493***
		(0.4524)	(0.4620)	(0.4911)
treat × yob = 2019		1.995**	2.083**	2.035**
		(0.8613)	(0.8848)	(0.8804)
p-value for nonzero pre-effect		0.772	0.799	0.414
Observations	15,352	15,352	15,352	15,352
R^2	0.16918	0.47434	0.50014	0.50506
Within R^2	0.00575	0.01144	0.04123	0.05067
treat fixed effects	✓	✓	✓	✓
post fixed effects	✓			
yob fixed effects		✓	✓	✓
parent_company fixed effects			✓	✓